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Lipids for Taste masking and Taste assessment in pharmaceutical formulations

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ABSTRACT

Pharmaceutical products often have drawbacks of unacceptable taste and palatability which makes it quite difficult for oral administration to some special populations like pediatrics and geriatrics. To curb this issue different approaches like coating, granulation, extrusion, inclusion complexation, ion-exchange resins, etc for taste masking are employed and among them use of lipids have drawn special attention of researchers. Lipids have a lower melting point which is ideal for incorporating drugs in some of these methods like hot-melt extrusion, melt granulation, spray drying/congealing and emulsification. Lipids play a significant role as a barrier to sustain the release of drugs and biocompatible nature of lipids increases their acceptability by the human body. Further, lipids provide vast opportunities of altering pharmacokinetics of the active ingredients by modulating release profiles. In taste sensors, also known as electronic tongue or e-tongue, lipids are used in preparing taste sensing membranes which are subsequently used in preparing taste sensors. Lipid membrane taste sensors have been widely used in assessing taste and palatability of pharmaceutical and food formulations. This review explores applications of lipids in masking the bitter taste in pharmaceutical formulations and significant role of lipids in evaluation of taste and palatability.

1. Introduction

Taste is a sensory response that results from a substrate receptor interaction (non-enzymatic) at the microvillous membrane level (Chandrashekar et al., 2000; Dulac, 2000). The local perturbations of the membrane result in electro-physiological and psychological events that lead to perception of taste. Local perturbations can also be affected by thermal stimulation of small areas of the tongue (Cruz and Green, 2000). Humans can discriminate among four primary tastes: sweet, sour, bitter, and salt. Bitter substances are tasted on the back of the tongue and palate, sour along the edges and palate, sweet at the tip and salt on the dorsum anterior. All these four modalities of the taste can be perceived by pharnyx and epiglottis. A fifth taste, umami has come to be recognized as an independent sensation. This taste is common in the Asian cuisines and is the taste of monosodium glutamate (Chaudhari et al., 2000; Firestein, 2000). Sweet, bitter and umami tastes are thought to be detected by G-protein coupled receptors that have the ability to bind with the tastant molecule (Mattes, 2011, 2009). Sour and salt tastes are due to high content of ions and protons in the food and their sensation arises due to transduction of sensitive cation channels (Ugawa et al., 1998). Taste sensation provides immediate sampling of the ionic, caloric, and potentially hazardous properties of the ingested food, beverage or chemical. Bitter taste perception provides protection against ingestion of poisonous compounds. Most of the medicines have a terribly bitter taste – so much that patient compliance is compromised. Drugs belonging to beta-lactam antibiotics, sulfonamides, steroids, macrolides, protease inhibitors, and anticancer drugs are major groups of bitter drugs.

Palatability is a supreme criterion for oral dosage forms (ODFs) especially for pediatric patients. Bitterness suppression has greatly helped in reducing worries of parents and physicians for administration of oral medicines to children thereby increasing the compliance to drug therapy. Consequently taste masked medicines are the preferred choice of not only parents and children but also of physicians. Increased demand of taste masked formulations in the market has motivated pharmaceutical companies for discovery and development in the field of bitterness reduction approaches. Most of these approaches are patented technologies used synergistically to control either the release of medication or altering the physiology of perception of bitterness. Different techniques are applied to mask bitter taste of active ingredients in

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pharmaceutical and nutraceutical formulations. Regulatory bodies like European Medicines Agency (EMA) and US Food and Drug Administration (US FDA) and International Conference on Harmonisation (ICH) have emphasized the role of palatability and taste masking in developing pediatric and other oral drug products. Several approaches of taste masking like film coating, complexation with cyclodextrin, using sweeteners or flavoring agents, ion-exchange resins, using lipids, etc. have been reported in literature (Walsh et al., 2014). Lipids have found their extensive use as functional excipients in some of these taste masking approaches. Lipids play a pivotal role in controlling release of drugsin the oral cavity thereby preventing buildup of a concentration higher than the bitterness threshold. Lipids have also been used for their outstanding effect in controlled release, moisture protection, revamping the swallowability and stability. In the 1960s, the lipids were explored for the first time in developing sustained release drug products (Schwartz et al., 1968; Siepmann et al., 2019). Thereafter, methods like coating, hot-melt extrusion, spray congealing were used for incorporating lipids in pharmaceutical formulations. Clarithromycin is a well-known macrolide antibiotic having bitter taste and for this reason it is extremely difficult to administer oral formulations of clarithromycin in pediatric patients. The bitter taste of this drug was successfully masked using glyceryl monostearate. A wax spherical matrix of clarithromycin containing 10 % aminoalkyl methacrylate copolymer E, 30 % clarithromycin along with 60 % glyceryl monostearate was optimized for suppressing bitterness (Yajima et al., 1999). Lipids also have a key role in revamping bioavailability of drugs and absorption. In lieu of the advantage of this excipient, it is very much susceptible to polymorphic changes and therefore stability issue is a downside of this excipient. This issue of polymorphic change and stability may be addressed with strategies like tempering during processing, tempering after processing, template effect, avoidance of melting, etc. (Becker et al., 2015; Vaassen et al., 2012).

Taste assessment trials are conducted with an aim to evaluate taste of tastants (food, chemical, drug etc.) and involve estimation of gustatory sensation responses in healthy human volunteers within well controlled procedures. Taste assessment trials are the standard and so far preferred method of taste assessment. Several in vitro taste assessment approaches have emerged as subsidiary methods for assessing taste characteristics of drug and drug products (Anand et al., 2007). These approaches include drug release studies utilizing taste sensors, specially designed apparatus and drug release by modified pharmacopoeial methods. Among these novel in vitro taste assessment apparatus and methodologies, taste sensors, also referred to as electronic tongue, is a preferred method for high-throughput taste screening and quality control of oral pharmaceuticals. Artificial lipids have been used in preparing taste sensing membranes for their use in developing taste sensors which are referred to as Lipid Membrane Taste Sensors (LMTS). LMTS have grown from their use as sensors in research to fabricate commercial taste sensing systems. These taste sensing systems are now available globally for characterizing and optimizing pharmaceutical/food formulations for their palatability.

This review article tries to present a comprehensive view on the use of lipids for their use as taste masking agent in pharmaceutical/food formulations and their other functional role in taste masked formulations. Further, lipids in fabricating taste sensing membranes have been discussed with their applications in taste masked formulation development and taste assessment.

2. Lipids for taste masking

Lipids are important excipients used extensively in pharmaceutical formulations and developing various drug delivery systems like liposomes, phytosome, ethosomes, solid lipid nanoparticles, and microparticles. As excipients, lipids play a pivotal role in modulating temporal and spatial aspects of drug targeting. Lipids have found their applications in modified release, moisture protection and improving swallowability. Apart from these applications, lipids are very much effective in masking bitter taste in oral formulations. It is reported that glyceryl distearate (Precirol® ATO 5) is a well-known lipid based excipient for masking the bitter taste, of course its efficiency in doing that depends on the drug and the amount of the lipid (Rosiaux et al., 2018; Lopes et al., 2017a, 2017b). Many researchers have reported the use of lipids for masking the taste of bitter drugs.

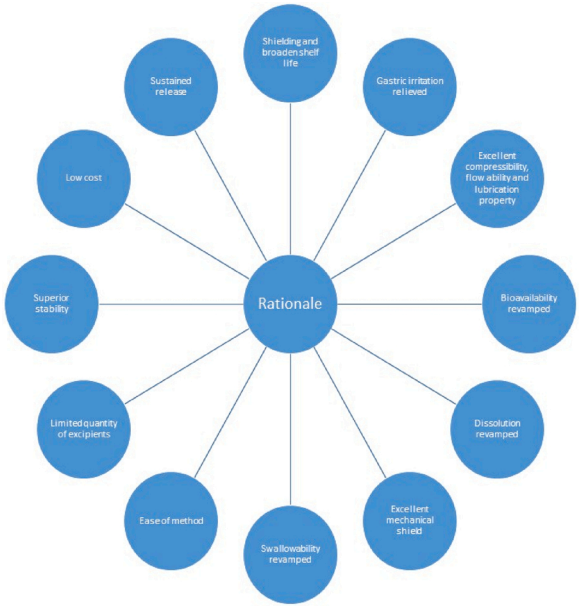


Fig. 1. Rationale for using lipids for taste masking in pharmaceutical and nutraceutical formulations.

2.1. Rationale for using lipids for taste masking

Lipids have proved themselves as an efficient taste masking agent and have drawn a special attention of the researchers for masking bitter taste of ODFs due to the following reasons.

a) Lipids have the ability to sustain release of bitter drugs in saliva resulting in an efficient control on the amount of drug which reaches to bitter taste receptors. With an optimized amount of lipids it is very much feasible to sustain release of the drug to such an extent that it remains below bitterness threshold without any appreciable decrease in the bioavailability of the drug which is an import consideration while designing immediate release drug products (Lopes et al., 2017a, 2017b; Dandagi et al., 2014; Kharb et al., 2014; Vaassen et al., 2012; Fini et al., 2008).
b) For sustained release taste masked oral drug products, lipids offer the additional advantage of sustaining the release of the drug without using any other polymer/excipients.
c) Lipids can revamp the bioavailability along with the taste masking of those bitter tasted drugs which are highly water soluble or hydrophobic in nature(Porter et al., 2007; Kalepu et al., 2013).
d) Lipids alter the dissolution profile of the bitter tasted drugs (Kawabata et al., 2015; Nabavi et al., 2015).
e) Lipids reduce gastric irritation in the stomach(Dubey et al., 2009; Sohi et al., 2004).
f) Some lipids ensure excellent compressibility, flow property and lubrication property for ODFs (Singh et al., 2010).
g) Swallowability of ODFs can be improved (Hauss, 2007; Singh et al., 2010).
h) The hot melt extrusion process and other melting methods for coating offers the advantage of a solvent free processing(Walsh et al., 2014; Khor et al., 2017; Patil et al., 2011).

Table 1
Classification of lipids with brief details.

Categories of Lipids	Brief details	Subcategories	Example of main Category
Fatty acyls	Building blocks of most of the lipids	Fatty acids and conjugates Hydrocarbons Oxygenated hydrocarbons	Dodecanoic acid, cyclopentenyl fatty acids, lipoic acid etc.
Glycerolipids	Metabolic fuels, membrane elements and signaling molecules	Monoradylglycerols Diradylglycerols Triradylglycerols	1-dodecanoyl-*sn*-glycerol, 1-hexadecanoyl-2 (9Z-octadecenoyl)-*sn*-glyceroletc.
Glycerophospholipids	Major constituents of the cell's lipid bilayer	Glycerophosphocholines Glycerophosphoethanolamines Glycerophosphoglucose lipids Oxidized glycerophospholipids	1-hexadecanoyl-2-(9Z-octadecenoyl)-*sn*-glycero-3-phosphocholineetc.
Sphingolipids	The sphingoid base analogues block the sphingolipids or play the role of surrogates of these.	Phosphosphingolipids Phosphonosphingolipids Neutral glycosphingolipids	Sphinganine, sphing-4-enine, 4-hydroxysphinganine, etc.
Sterol lipids	Major role as membrane lipids of the cells	Sterols Steroids Secosteroids Bile acids and derivatives Steroid conjugates Hopanoids Others	Cholesterol, cholest-5-en-3β-yl-dodecanoate, β-estradiol, androsterone, cortisol, vitamin D_2 etc.
Prenol lipids	The chemical structure is mainly based on the isoprenoid units andare formed by mevalonic acid pathway or by methylerythritolphosphate pathway.	Isoprenoids Quinones and hydroquinones	2E-geraniol, dimethylallyl pyrophosphate, retinol, etc.
Saccharolipids	Fatty acids are conjugated with the sugar backbone andcompatible with bilayers of the membrane.	Acylaminosugars Acylaminosugar glycans	Diacylaminosugars, tetraacylaminosugars, etc.
Polyketides	These are originated by enzymes especially multimodular enzymes. Their skeleton is adapted by oxidation, glycosylation, hydroxylation etc. reactions.	Macrolide polyketides Aromatic polyketides Nonribosomal peptide/polyketide hybrids Others	Epothilone D, erythromycins, tetracylines, nystatins, avermectins,griseorhodin A, 6-deoxyerythronolide B etc.

i) Generally small amounts of lipid excipients are employed when compared to other excipients (Zheng et al., 2018).
j) Lipids offer superior stability to ODFs (Zheng et al., 2018).
k) Due to low price of lipids, the entire cost of production is reduced (Zheng et al., 2018).

Fig. 1 depicts the above-mentioned summarized points as rationale for using lipids in pharmaceutical and nutraceutical formulations.

2.2. Classification of lipids

Lipids can be classified in many ways and several classifications schemes have been reported in scientific literature. Lipids have been classified in three broad categories, viz. (1) simple lipids (like waxes and triacylglycerol etc.), (2) conjugated or complex lipids (like glycolipids, phospholipids and glyceroglycolipids, etc.) and (3) derived lipids (like terpenes, steroids and fatty acids etc.) (Gyamfi et al., 2019; Hatziantonioy and Demetzos, 2008). The International Union of Pure and Applied Chemistry (IUPAC) has classified lipids on the basis of position of the double bond with respect to the carboxyl carbon (−COOH), e.g.linoleic acid can be considered according to this classification as Δ9,12−18:2 or cys, cys-9,12−18:2. Lipids havea lso been classified by the position of the double bond with respect to the terminal methyl ($-CH_3$) group of the fatty acids. In this case ω or n naming system is applied, i.e. in this case linoleic acid can be considered as 18:2 ω6 or 18:2 n-6, where ω or n represents the carbon number from the carbon of the first$-CH_3$ group (Kerr et al., 2015; Gidez, 1984). In one other classification system of lipids have been proposed, lipids have been divided into six different categories, given and briefly discussed in the Table 1 (Fahy et al., 2011, 2005).

2.3. Pros and cons of using lipids for taste masking

Lipids are expedient in many cases in comparison to other materials for their use as taste masking agents. Some major precedence of lipids over the other taste masking agents are sorted below.

(a) Acceptable taste and good flavor governs patient acceptance and popularity of ODFs. For preparing tablet compression is essentially required, except for 3D printing and freeze drying methods of shaping tablets. Hence. it is very important to incorporate excipients which can withstand high pressures during compression. Lipids are renowned plastic components which can withstand the compression force during the tableting process. Table 3 describes different works on taste masking of bitter tasted agents with lipids and Table 4 describes some patents granted in different countries on these works (Iswandana et al., 2018; Petrovick et al., 2017; Eckert et al., 2014; Fini et al., 2008; Duru et al., 2004; Suzuki et al., 2004).
(b) Being hydrophobic in nature and lipids are easily digestedin the aqueousenvironment of the human body and absorbed easily. Glycerol and fatty acids combine to form triacylglycerols, which is having ester bonds in their chemical structure. These bonds are hydrolysed in the presence of water and monoacylglycerol leading to formation offree fatty acids, which are highly water soluble in nature leading to dissolution and subsequent absorption in the human body (Corzo-Martínez et al., 2020; Calvo-lerma et al., 2018; Gonçalves and Cyrino, 2014; Merican and Shim, 1995).
(c) It is a very noteworthy point that lipids are considered as generally recognized as safe (GRAS) (U.S. Food and Drug Administration, 2016, 2004). GRAS status allows the use of lipids in pharmaceutical formulations.
(d) In case of extended release ODFs dose dumping is a serious issue and among various reasons for dose dumping alcohol induced dose dumping is a very recurrent issue. Alcohol induces quick release of the drugfrom extended release formulation leading to failure of extended release ODFs and subsequent issues of adverse effects and toxicity. With lipid based extended release ODFs this serious issue can be curbed due to insolubility of lipids in ethanol (Maboos et al., 2018; Uronnachi et al., 2014; Walsh et al., 2014). However, in one such paper, interaction between alcohol and pellets has been reported to change drug release patterns from pellets. In this case paracetamol or codeine phosphate were taken

as active agents along with Compritol® and Precirol®which were added in the matrix and pellets were prepared using hot-melt extrusion method. However, alcohol induced dose dumping was observed from the matrix formulation (Jedinger et al., 2015).

(e) Lipids are generally more compatible with most of the drugs and other excipients in the pharmaceutical formulations (Silva et al., 2015; Kumar et al., 2013). It is noteworthy that apart from the taste masking, in general lipids have demonstratedadequatecompatibility with active agents and other excipients in different formulations.

Despite having several advantages there are some negative sides of lipids. Several reports havefocused on the drawbacks of lipids along with the precedence of these. Among these some major drawbacks are as follows:

(a) Lipids like triglycerides and fatty acids having long chain incresethe time of residence of drug delivery system in the gastrointestinal tract (GIT). Ingestion of fatty foods with drug delivery system may have a synergistric effect on the residence time.Triglycerides interact with drugs and the transportation of lymph through intestine is elevated leading to modification of disposition. Such changes in pharmacokinetics and subsequently pharmacodynamics of the drug may create serious safety and toxicity issues especially for those drugs having narrow therapeutic index (N'Goma et al., 2012; Gershkovich and Hoffman, 2007; Wagner et al., 2001).

(b) For masking the bitter taste of the active agent, it should be well coated with lipid material and for this it is very much essential to solubilize the drugcompletely in lipid material. It is observed that in many cases active agents are not soluble in lipid materials which restrict application of this method. Various physicochemical parameters, hydrophilic-lipophilic balance (HLB) value, partition coefficient, etc,mayinfluence the solubility (Nanjwade et al., 2011).

(c) Formulating drugs with high melting point lipids may significantly reduce absorption of the drugs leading to reduction in bioavailability which may cost therapeutic efficacy. For better absorption generally lipids having low melting points are chosen. Particularly in this regard sometimes the cost becomes a crucialfactor while determining the lipid of choice (Nanjwade et al., 2011).

2.4. Selection of lipids for taste masking

Superior efficiency in taste masking and swallowability of the bitter tasted agents is one of the key factors in optimizing ODFs. Several types of lipids are available for bitterness suppression but only some are suitable. Melting point, charge and stability of lipids are some primary determinants for their selection.

2.4.1. Melting point of lipids

Among lipids fatty acids are well-known for their use in taste masking. Fatty acids have different chain length and the degree of unsaturation on which melting point of a lipid depends. This unsaturationcan be observed in the alkyl chains of lipids as the presence of one or more double bonds.In case of polyunsaturated fatty acids (PUFAs) one methylene group (−CH2) is there between two double bonds. Lower melting points denote the unsaturation of fatty acids and high melting pointare observed for saturated fatty acids. Due to unsaturation PUFAs of octadecanoic acid (C_{18}) have lower melting point of 13.4 °C. Stearic acid, a long chain saturated fatty acid has a melting point of 69.6 °C while oleic acid, a monounsaturated fatty acid,melts at 13.4 °C (Berg et al., 2002). Melting points helps in selecting the methods through which fatty acids may be incorporated into the formulation. Low melting point makes them suitable for a heat free process of coating or

Table 2
Physical characteristics along with their USFDA approved maximum amount per unit dose of some lipids.

Lipid	Melting point/ range	Degree of unsaturation	Maximum amount per unit dose			Reference
			Capsule	Tablet	Powder	
Glyceryl monostearate	74 °C	1	1.75 mg (delayed release capsule); 32.4 mg (extended release capsule)	1.7 mg (delayed release tablet); 154 mg (extended release tablet); 264.3 mg (film coated extended release tablet); 7.5 mg (orally disintegrating delayed release tablet)	Not listed	(National Center for Biotechnology Information, 2020a, 2020b, 2020c, 2020d; U.S. Food and Drug Administration, 2020a)
Lecithin	236–237 °C (soybean lecithin)	3.5	Not listed	0.16 mg (delayed release tablet); 10 mg (extended release tablet); 0.35 mg (film coated extended release tablet)	167 mg/ 5 mL (powder for suspension)	(National Center for Biotechnology Information, 2020a; U.S. Food and Drug Administration, 2020a)
Carnauba wax	82–86 °C	Unspecified molecular formula	80.67 mg (extended release capsule)	230 mg (delayed release tablet); 300 mg (extended release tablet); 140 mg (film coated extended release tablet); 0.11 mg (multilayer extended release tablet)	Not listed	(Khor et al., 2017; U.S. Food and Drug Administration, 2020a)
Glyceryl dibehenate	72.3 °C	0	3.5 mg (delayed release capsule); 0.5 mg (extended release capsule)	12.5 mg (delayed release tablet); 142.5 mg (extended release tablet); 10 mg (film coated extended release tablet); 5.6 mg (orally disintegrating tablet)	0.08 mg/mg (powder for suspension)	(Deshpande et al., 2017; U.S. Food and Drug Administration, 2020a)
Glyceryl tristearate	55 °C (α), 64.5 °C (β'), 73 °C (β)	3	225 mg (capsule)	2.3 mg (chewable tablet)	Not listed	(National Center for Biotechnology Information, 2020b; U.S. Food and Drug Administration, 2020a
Stearic acid	68.8 °C	1	25.5 mg (extended release capsule)	30.84 mg (delayed release tablet); 187.5 mg (extended release tablet); 9 mg (film coated extended release tablet); 10 mg (orally disintegrating)	1203 mg/ 5 mL (powder for suspension)	(National Center for Biotechnology Information, 2020b; U.S. Food and Drug Administration, 2020a

Table 3
Hot-melt extrusion method used for taste masking by lipids in research woks at a glance.

Lipid	Active Ingredient	Other Important Excipients	Method/Technique	Dosage Form/ Delivery System	Significant Outcome	Reference
Hard fat (Witocan® 42/44 mikrofein), glyceryl distearate (Precirol® ATO 5), glyceryl trimyristate (Dynasan® 114)	Metformin hydrochloride		Solid lipid extrusion/ spheronization	Pellets, ODTs	Compression of lipid (content about 15 %) pellets into ODTs did not adversely affect taste masking efficiency and drug release profiles.	(Petrovick et al., 2017)
Carnauba wax	Quercetin	Shellac, zein	Hot melt extrusion microencapsulation	Microcapsules	Carnuaba wax-microencapsulated drug was found superior in taste masking when assessed by in vitro drug release and LMTS.	(Khor et al., 2017)
Glyceryl monostearate (Imwitor® 491)	Praziquantel	Eudragit® E PO	Extrusion and spray drying		Drug-lipid and Eudragit® E PO extruded particulates successfully entrapped crystalline drug and masked bitter taste.	(Münster et al., 2017a)
Lecithin	Curcumin C3 Complex (curcumin, demethoxy curcumin, bisdemethoxy curcumin)	HPMC, Iiomalt	Hot melt extrusion	Amorphous solid dispersion	Amorphous solid dispersion increased bioavailability to 13 fold and improvements in pharmacokinetics of active ingredient	(Chuah et al., 2014)
Compritol®, Precirol® and white beeswax	Quetiapine fumarate	Mannitol	Hot melt extrusion, freeze drying	Pellets, ODTs	Precirol®/beeswax/ quetiapine in ratio of 1:1:1 was found optimum for zero order stable drug release for 12 months without any burst release.	(Refaat et al., 2016)
Hard fat (Witocan® 42/44 mikrofein), glyceryl distearate (Precirol® ATO 5), glyceryl trimyristate (Dynasan® 114)	Metformin HCl		Solvent-free cold extrusion/ spheronization	Pellets	An On-line and in-line drug release assessment of pellets with electronic tongue revealed only 10 % drug release within 30 s from pellets containing 30 % lipids.	(Petrovick et al., 2016)
Hard fat (Witocan® 42/44 mikrofein), glycerol distearate (Precirol® ato 5) and glycerol trimyristate (Dynasan® 114)	NXP 1210	Polyvinylalcohol (PVA)–polyethyleneglycol (PEG)-graft copolymer (Kollicoat® IR) and crospovidone (Polyplasdone® Xl-10)	Solid lipid extrusion	Pellets	Immediate release pellets of drug capable of masking bitter taste by designing them to have a lag of drug release at initial time points.	(Vaassen et al., 2012)
Hard fat (Witocan® 42/44 mikrofein), glycerol distearate (Precirol® ato 5, glycerol trimyristate (Dynasan® 114), glyceryl tripalmitate (Dynasan® 116), glyceryl tristearate (Dynasan® 118) and, glyceryl monostearate (Imwitor® 491), powdered glyceryl dibehenate (Compritol® 888 ATO) and cetyl palmitate (Precifa®) were supported by Gattefossé	Praziquantel	Polyethylene glycol (PEG) 1500, 6000, colloidal silicium dioxide (Aerosil® 200)	Solid lipid extrusion	Powders	Solid lipid extrudates containing high loading of drug (70 %) and 20 % PEG were sufficiently taste masked as assessed in cats.	(Witzleb et al., 2011b)
Witocan 42/44 mikrofein (powdered hard fat) and Dynasan 114 (glyceryl trimyristate powder) from Sasol GmbH (Witten, Germany); Precirol ATO 5 (glyceryl palmitostearate powder) and Compritol 888 ATO (glyceryl dibehenate powder),	Benzoic acid	Carrageenan and microcrystalline cellulose	Extrusion and spheronisation	Pellets	Powdered lipids were used as binders for preparing pellets by extrusion-spheronisation process. However, no study on taste masking was performed.	(Krause, 2008)
Glyceryl behenate (Compritol 888 ATO)	Enrofloxacin	Colloidal silicium dioxide (Aerosil 200)	Solid lipid extrusion		Reduction in size of extrusion die (diameter) reduced drug release leading to taste masking.	(Michalk et al., 2008)

complexation.

2.4.2. Stability of lipids

Generally, if the degree of unsaturation of a lipid in acyl chains is more, it is very likely to be degraded by oxidation. Saturated lipids have shown superiority over the prior compounds when stability is concerned. Despite having good stability, saturated lipids have critical formulation issues like high melting point. Unsaturated lipids may be solubilized by lowering the degree of unsaturation. Oleic acid is a monounsaturated fatty acid which is more stable when compared to PUFAs. On the other hand, lipids are also highly prone to hydrolysis. Hydrolysisdependson the degree of unsaturation of lipid and some environmental factors like pH, temperature, ionic strength, etc. Lipids having optimum melting point and lower degree of unsaturation are preferred in formulating taste masked dosage forms (Yadav et al., 2011; Grit et al., 1989; Kepsil and Dennis, 1981).

2.4.3. Charge on lipids

The complexation or the coating of lipid on the bitter tasted active agents depends on the charge. Most of the lipids are negatively charged because of having −COOH group at the terminal end of their chemical structure. Phosphatidic acid, a renowned taste masking lipid belonging to phospholipid group, is a negative charged lipid (Pöyry and Vattulainen, 2016; Fini et al., 2008; Nakamura et al., 2002; Katsuragi and Kurihara, 1993a, 1993b). In many cases when a single lipid can not show the desired taste masking effect, two or more lipids are mixed and used for suppressing bitterness. Apart from this ratio of unsaturation and respective concentrations of lipids are also subjected for a parameter in this case. In many cases it is observed that the stability is elevated in case of the mixture rather than that of the single lipid, which is due to the fact of formation of strong bond between two lipids (Han and Zhang, 2019; Tang et al., 2017; Suzuki et al., 2004; Takagi et al., 2001; Katsuragi et al., 1996).

Melting points and degree of unsaturation of some lipids along with their maximum amount per unit dose (USFDA approved) are given in Table 2. Degree of unsaturation is calculated by the following formula (Badertscher et al., 2001).

$$\text{Degree of unsaturation} = 1/2\ (2 + 2a - b + c–e)W$$

here a. b, c and e represent the number of carbons, hydrogen, nitrogen and halogen atoms respectively.

2.5. Methods and technologies for taste masking with lipids

Processing methods are crucial when preparing pleasant tasting formulations. Hot melt extrusion, spray drying/congealing and melt granulation are some key methods which have been used for preparing taste masked formulations containing lipids.

2.5.1. Hot-melt extrusion/coating methods

Hot-melt extrusion (HME) is a process of transforming a mixture of powder into a uniform shaped compact productthrough a hot barrel maintained at a high temperature and pressure (Maniruzzaman et al., 2012; Lang et al., 2014; Repka et al., 2008).The mixture of drug, polymers/lipids, plasticizers, thermal lubricants and other excipients are converted to a melt at high temperature and this melt is forced through extruder, subsequently mixed with other substances, cooled to obtain materials with designated properties. The melting point of the lipid and temperature of HME process play a pivotal role as uniform molecular level mixing is feasible above melting point which may with the choice of lipid and other excipients used in the process (Crowley et al., 2007). Extrudates of designated properties are often milled and finally mixed with other excipients to transform into desired dosage forms. In HME process selection of suitable ingredients, quantitative optimization of the composition and optimization of critical process parameters are required on a case to case basis where Design of Experiment (DoE) coupled with Process Analytical Technology (PAT) has proven roles. Twin-screw extruder and single screw extruder are two techniques of HME where barrel temperature, feed rate, mixing speed, melt pressure, screw speed and motor load are critical process parameters (Repka et al., 2007; Breitenbach, 2002).

HME process does not require any solvent for dissolving drug and other components of the formulation. HME, being a green technology, offers the advantages of reducing chemical degradation of the drug and avoids toxicity of residual organic solvents. Further, it also eliminates the need of additional processing steps for removing solvent from the finished product. Due to few processing steps and continuous operation, HME is easy to be scaled up from a lab to pilot scale and subsequently to industrial scale. It can be used for preparing a huge range of dosage forms and drug delivery systems across various routes of administration. Additionally it can address issues like poor bioavailability, modify release and suppression of bitter taste. However, this method is not useful for thermolabile drugs, materials which cannot withstand high shear stresses and feedstock with poor flow properties (Lang et al., 2014; Crowley et al., 2007). Due to low melting point of lipids, HME with lipids is a method of choice when dealing with thermolabile drugs or substances. On some occasions lipids also eliminate the need to use any plasticizer as they soften sometimes below the melting point and thereby allowing extrusion.

HME is a widely used method for masking taste of bitter drugs where the extrudates of lipids and/or polymers are fabricated. HME process was used to prepare taste masked microencapsulated quercetin powders (Khor et al., 2017). Preheated carnauba wax (t_m 85 °C, preheat temperature 80 °C)in different weight ratio (50/50, 30/70, 40/60) with quercetin was plasticized with 20 % w/w water in a turbula mixer and fed to a twin screw extruder at a speed of 100 rpm without using any die. Collected extrudates were cooled and subjected to reduction is size in a bead mill to obtain carnuba wax microencapsulated quercetin powders. A significant reduction in dissolution rate of quercetin, with just 3.14 ± 0.23 % of quercetin release, was achieved in simulated salivary pH 6.8 medium within 1 min from microencapsulated powders of quercetin. Bitterness sensory output (0.6 mV) as assessed from a taste sensor (TS-5000Z) for carnauba wax-microencapsulated quercetin was much less than the bitterness sensory out which a human tongue can detect (1 mV).In one another study, hot melt coating of bees wax and cetyl alcohol was applied onto pellets of two bitter drugs, bromhexine hydrochloride and salbutamol sulphate, for masking taste (Patil et al., 2011). Extrusion spheronisation was used to prepare pellets of bromhexine hydrochloride and salbutamol sulphate with MCC, lactose and povidone. A 2–3 % coating of pellets was successfully demonstrated to achieve acceptable taste masking when assessed through human sensory taste panel. Further, drug release from pellets in first five minute was less than bitterness threshold concentration (300 μg/mL). Apart from these, the bitter taste of quetiapine fumarate was masked by applying Compritol®, Precirol® and white beeswax through a HME technique (Refaat et al., 2016). Melts of lipid carriers, alone or in combinations, were achieved at 90 °C on a heated water bath, allowed to congeal at about 30 °C with stirring followed by extrusion through a sieve of 1.25 mm aperture size. Extruded rods were reduced in size to through a sieve of 0.76 mm aperture to obtain micro pellets of size 0.5−1 mm in size. These pellets were subsequently used to prepare orally disintegrating tablets of acceptable taste.

High daily dose of curcumin (2−12 g) limits the use of curcumin in food products due to strong taste, color and odor. Melt extrusion at different temperature in range 50−150 °C, 0.75 kg/h feed rate, screw aperture 3 mm, and screw speed of 200 rpm in a co-rotating twin screw extruder was employed for a powder mixture comprising curcumin (10 %) and a matrix of lecithin (10 %) HPMC (75 %) and isomalt (5%) (Chuah et al., 2014). Amorphous solid dispersions of curcumin were achieved for addressing issues of bioavailability and masking strong odour and taste of curcumin.

A cold solvent free extrusion technique was developed for preparing sodium benzoate microcapsules with different mixtures of glyceryl tristeate (Precirol® ATO5), stearic acid and hard fat (Witocanw 42/44 mikrofein) (Breitkreutz et al., 2003). Lipids and drug were mixed in a mixer followed by extrusion through a rotary ring die press to obtain cylindrical extrudates of length 1−3 cm and a diameter of 1 mm. Spheronizer was used to prepare microparticles from cylindrical extrudates. Obtained microspheres were coated with Eudragit E at 8% polymer mass level to achieve acceptable taste masking. In another study, cold solvent free extrusion technique was also applied to various binary, ternary and quaternary mixtures of lipids to prepare immediate release pellets of sodium benzoate for pediatrics (Krause, 2008). A formulation comprising glycerol trimyristate, glycerol distearate and hard fat gave good pellets characterized by narrow size distribution, low aspect ratio and complete drug release.

Effect of die diameter (0.3 to 0.5 mm) and temperature (high or low to melting point of glyceryl benenate) was studied on extrudates consisting of enrofloxacin (50 %), glyceryl behenate (49 %) and 1% colloidal silicium dioxide (Michalk et al., 2008). No significant different in drug release was observed with temperature change with same diameter extrudates. However, drug release profiles were found dependant on the specific surface area of milled particles which depends on extrusion diameters. Dissolution test for a short time (15 s to 1 min) revealed for milled extrudates versus milled precipitates revealed higher drug release from milled precipitate due to high surface area and agglomeration of particles. Drug release from extrudates was relatively slower and capable of masking taste of drug. Increase in extrusion diameter (0.3 mm to 0.4 mm and to 0.5 mm) resulted in subsequent increase in surface area of extrudates leading to faster dissolution in a short time (15 s). Thereby a control on drug release was achieved with small die diameter which was due to the use of small die which produced extrudates characterized by low surface area.

A continuous extrusion process including post extrusion milling of extrudates highlighted the need for the addition of antistatic agent (about 10 % of liquid PEG) to prevent electrostatic charging of extrudates (Witzleb et al., 2011a). Further, in this study various pellet formulations of praziquantel were prepared with lipidic excipients like glyceryl tripalmitate, glyceryl tristearate, glyceryl monostearate, solid paraffin, glyceryl dibehenate and cetyl palmitate. Silicium dioxide was used for improving the flow properties and dosing precision. High (70 %) loading of praziquantel in extrudates of size down to 0.2 mm and the addition of 20 % PEG produced acceptable taste masked extrudates. Pellet formulations prepared from small diameter extrudates were added to dry and canned foods and assessed for their palatability in cats. Small diameter extrudates, the addition of PEG and the presence of water in canned food did not affect taste masking efficiency.

Solid lipid extrusion method was used to achieve taste masking in immediate release formulations of praziqantel (Münster et al., 2017a). A mixture of praziquantel, glyceryl monostearate and Eudragit E PO in ratios 1:1:0.3 and 1:1:0.6 by weight was extruded at temperatures below and above the melting point of glyceryl monostearate (melting point range: 65−80 °C, operating temperatures 50 °C or 70 °C). Glyceryl monostearate was added to enhance taste masking efficiency of Eudragit EPO and to increase flow properties of the mixture for facilitating a constant feeding during extrusion. Taste masking efficiency of prepared formulations was assessed through an in vitro drug release test in simulated salivary fluid for upto 5 min. The drug release in simulated salivary fluid from of both drug loads was 0.01 mg/mL after 60 s which was less than the bitterness threshold of 0.03 mg/mL from prepared extrudates. However, delayed release of drug was slightly higher, i.e. 0.06 mL/min, after 5 min.

Acceptable taste masking and fast and complete drug release was achieved for a BCS Class II drug NXP 1210 from pellets comprising extrudates comprising mixture of lipids (Vaassen et al., 2012). Glycerol distearate, hard fat and glycerol trimyristate in different compositions were used to prepare extrudates. Taste masking was assessed as a drug release less than 5% within 2 min. Partial replacement of hard fat in extrudates with glycerol trimyristate or glycerol distearate increased drug release. A formulation comprising 15 % of hard fat, 2.5 % of glycerol distearate and 2.5 % of glycerol trimyristate was found suitable for extrusion at room temperature and subsequently pelletization with 50 % of NXP 1210 and 30 % of crospovidone and Kollicoat® IR (PVA–PEG-graft copolymer).

Some literature reports on taste masking with lipids through HME process are presented in Table 3.

2.5.2. *Spray drying/coating methods*

Spray-drying is a process wherein the slurry is dried using hot gas/air to get dry powder(Li et al., 2019). In this process the active agent, lipid/polymer and other excipients are taken as the feed which are introduced by spraying using an atomizer.The solution is sprayed by atomizer to the chamber containing hot gas/air to evaporate the water or organic solvent. This is another well-known method for taste masking, where the coating of lipid serves as the physical barrier preventing release of drug. Solvent, both aqueous or non aqueous, have been used for dissolving/dispersing drug, polymer/lipids and other excipients. Temperature,pressure, air flow patterns, fill volume, drug-excipient ratio, particle size and particle size distribution are some critical processing parameters which require careful optimisation (Yang, 2020; Sollohub et al., 2011; Bora et al., 2008).Spray drying method is an immensely popular method of preparing free flowing powders in pharmaceutical industry as itis an inexpensive, well established, straightforward and easy to scale up method. Different spray drying systems available are open cycle systems, closed cycle systems and semi-closed cycle systems.

A significant advantage of the spray drying process is the consistency in the size distribution of powder particles. Apart from this, fast processing and the adaptability to coat particles for taste masking and modified release offers wide acceptance. Despite these advantages, this method has disadvantages like high consumption of energy, difficulty in obtaining uniqueness of particle size, less production rate and its applicability to only non-thermolabile drugs are limitations for cetain applications (Li et al., 2019).

A spray dried triple emulsion was used to encapsulate orange oil and control its release (Edris and Bergnsta, 2001). Orange oil oil-in-water emulsion (O_1/W) was prepared in a colloid initially with sodium caseinate as emulsifier and later homogenized in a high pressure homogenizer. Double emulsion (O_1/W/O_2) was prepared from oil-in-water emulsion with a second oil (rape seed oil) and a lipid emulsifier in a electric stirrer. In this process, Oil-in water emulsion, and lactose solution (14 %) was added drop by drop to a mixture of rape-seed oil and emulsifier at a slow stirring rate for 50 min. Triple emulsion (O_1/W/O_2/W) was achieved with further re-encapsulation of the double emulsion in a water phase wherein lactose monohydrate and sodium caseinate were also incorporated into the formulation. Finally, a spray drying method in a laboratory scale spray dryer was developed for drying the triple emulsion. A secondary coating was thus provided by spray drying for obtaining a powder with excellent flow properties and protection from deterioration.

A coating of glyceryl monostearate and Eudragit E PO was applied onto praziquantel via a spray drying process (Münster et al., 2017a). Spray dried powders were amorphous and resulted in a faster drug release when compared to an extrudate formulation with the same drug load. However, the acceptable taste masking could not be achieved with spray dried formulation as in vitro drug release offered a very faster build up of concentration greater than the bitter threshold concentration.

A pediatric formulation comprising safe excipients lecithin and sodium caseinate was developed through a spray drying process for masking the taste of acetaminophen (Thi et al., 2012). Astree electronic tongue and an indirect method based on drug release were used to assess taste masked formulations. To simulate the in vivo environment of the human mouth, a syringe pump was used with small flow rates and small

Table 4
Spray drying/congealing and fluidized bed coating methods used for taste masking by lipids in research woks at a glance.

Lipid	Active Ingredient	Other Important Excipients	Method/Technique	Dosage Form/ Delivery System	Significant Outcome	Reference
Precirol® ATO 5	Paracetamol	α- lactose monohydrate, MCC and PVP	Granules prepared by wet granulation were coated by hot melt coating in a modified fluidized bed system.	Coated granules, tablets	Hot melt coating of granules and subsequent tableting was suggested as a possible mean for achieving taste masking or prolonging release of the drug.	(Milanovic et al., 2020)
Octadecanol and glycerin monostearate	Ibuprofen		Spray congealing process with three different cooling methods water-cooling, air-cooling, and citric acid solution-cooling	Microspheres	Internal 3D structure and external morphology of the microspheres was studied with synchrotron radiation X-ray micro-computed tomography for their possible role in taste masking and drug release.	(Qin et al., 2019a, 2019b)
Stearic acid, cetyl alcohol, or cetyl esters	Paracetamol	Paraffin wax	Spray congealing	Microparticles	Microparticle size was found depended on viscosity of the viscosity of the melt. Cetyl alcohol and cetyl esters were found potential taste masking agents.	(Ouyang et al., 2019)
Stearic acid, cetyl alcohol, or cetyl esters	Paracetamol	Paraffin wax	Spray congealing	Microparticles	Wax-lipid blends produced small size microparticles characterized with high drug content, high surface loading and high drug release when compared to plain wax microparticles.	(Ouyang et al., 2018a, 2018b)
Tripalmitin (Dynasan® 116))	N-acetylcysteine	Polysorbate 65	Fluidized bed hot melt coating	Multiparticulate dosage forms for children, geriatrics and dysphagics	Hot-melt coating was applied for suppression of sour taste. An in vitro method for assessing taste was developed. DSC and XRD revealed stable β-polymorph of tripalmitin for maintaining a stable drug release over long periods.	(Becker et al., 2016b)
Vegetable fat	Proanthocyanidin-rich cinnamon extract		Using spray chilling technique	Solid lipid microparticles	Proanthocyanidin-rich cinnamon extract and vegetable fat ratio of 8:72 was capable of masking the bitter and astringent taste.	(Tulini et al., 2016)
Octadecanol	Paracetamol	Eudragit E100	Spray congealing	Microspheres	Synchrotron radiation X-ray computed microtomography was used to assess internal 3D structure of lipid microspheres for drug release and taste masking.	(Guo et al., 2016)
Tripalmitin	N-acetylcysteine	Polysorbate 65	Fluidized bed hot melt coating	Multiparticulates	ICH Q9 quality risk management guideline was applied for developing multiparticulates using a hot melt coating method.	(Stocker et al., 2017)
Stearic acid	Clindamycin HCl	Eudragit® EPO	Fluidized bed coating	Coated beads incorporatd into ODTs	Taste masking suspension formulation contained 1.8 % stearic acid and 12 % Eudragit EPO. Stearic acid forms a soluble salt with the Eudragit® PO.	(Cantor et al., 2015)
Lecithin	Paracetamol	Sodium caseinate and calcium caseinate	Spray drying	Spray dried powders	2^4 full factorial design was employed for optimizing the amount of lecithin, amount of caseinate and the spray rate for drug release within 2 min.DSC and SEM studies revealed proportion of lecithin and caseinate as a critical formulation parameter for achieving taste masking.	(Thi et al., 2013)
Lecithin	Paracetamol	Sodium caseinate	Spray drying	Spray dried powders	Higher concentration of lecithin in spray dried powders resulted in high efficiency of taste masking. With a ratio caseinate: lecithin of 5:1.5 ratio, the surface of particles was covered entirely with caseinate leading to successful taste masking which was also confirmed with electronic tongue assessment and in vitro drug release study.	(Thi et al., 2012)
	Paracetamol		Spray congealing	Wax matrix particles		

(*continued on next page*)

Table 4 (*continued*)

Lipid	Active Ingredient	Other Important Excipients	Method/Technique	Dosage Form/ Delivery System	Significant Outcome	Reference
Glyceryl monostearate		Aminoalkyl methacrylate copolymer E (Eudragit E)			3^3 Box–Behnken design was used to evaluate variables in a model. A new time-dependent diffusivity model, which assumes the change in the drug's diffusion coefficient, was proposed to predict the drug release from wax particulates. Multiple regression analysis was applied to formulate acetaminophen to achieve bitterness suppression without compromising on bioavailability.	(Nitanai et al., 2012)
Stearic acid, carnauba wax, lecithin	Mesalazine	Eudragit® L100, mannitol	Spray congealing	Layered lipid microcapsules	pH dependent lipid microparticualte product comprising lipid (carnauba wax/ Eudragit L100) cores layered with spray dried mannitol/ lecithin was fabricated for colonic drug delivery.	(Balducci et al., 2011)
Palmitic acid and stearic acid	Benzoic acid		Spray chilling method	Microspheres	XRD and thermal studies on binary 1:1 stearic acid and palmitic acid based microspheres revealed significant interactions, both on the surface and deeper in the core, of microspheres with alkaline buffers due to formation of acid soaps.	(Qi et al., 2008)
Stearic acid or dibutyl sebacate	Quinine sulphate	Eudragit E PO	Extrusion-spheronisation and coating (bottom spray wurster fluid bed process)	Coated pellets for flexible pediatric dosing	Lipids were used as plasticizer in the taste masking coating composition.	(Kayumba et al., 2007)
Palmitic acid, stearic acid, sodium palmitate and sodium stearate			Spray chilling method	Microspheres	DSC and XRD analysis revealed changes in structure of fatty acid microspheres when exposed to alkaline media. Acid-soap formation was proposed as a mechanism for these structural changes and resulting changes in the drug release profiles.	(Qi et al., 2006)
Glyceryl monostearate	Clarithromycin	Aminoalkylmethyl methacrylate copolymer E (Eudragit E)	Spray congealing agglomeration	Wax matrix disc	Optimum processing conditions of spray congealing process for facilitating formation of beta form of glyceryl myristate were identified.	(Yajima et al., 2003)
Stearic acid	Cefuroxime axetil		Spray chilling	Microspheres	The drug release from cefuroxime axetil (15 %) and stearic acid (85 %) based microspheres differs significantly in dissolution media of different pH. Prepared microparticles satisfied the requirements of taste masking and immediate drug release in GIT.	(Robson et al., 1999, 2000a)
Glyceryl monostearate	Clarithromycin	Aminoalkylmethyl methacrylate copolymer E (Eudragit E)	Spray congealing agglomeration	Wax matrix	Atomizer wheel speed and liquid feed rate for a spray congealing process was optimized for masking the bitter taste of drug.	(Yajima et al., 1999)
Hydrogenated oil (Lubri wax 103), sucrose fatty acid ester (Ryoto Sugar Ester S-370 F), stearic acid	Indeloxazine HCl	Pluronic F-68, lactose, mannitol	Fluidized bed granulator-coater (side-spray method)	Wax matrix microparticles	Developed microparticles were successful in reducing the release of drug to a level less than the bitterness threshold of the drug.	(Sugao et al., 1998)

volumes of the aqueous release medium. Lecithin to caseinate ratio of 0.5:1 was found optimum for covering the surface of acetaminophen and to curtail the release of acetaminophen below bitterness threshold. In a different study by the same group, a 2^4 full factorial optimization was employed for investigating the effect of amounts of lecithin (0.5 and 1.5 %) and calcium caseinate (1–5 %) and the impact of spray flow (300 L/h and 60 L/h) and temperature (130 °C and 150 °C) on the drug release within 2 min from spray dried formulations of acetaminophen (Thi et al., 2013). Lecithin was used to control the release of drug from the developed formulations below bitterness threshold of 0.21 mg/mL to mask the bitterness.The properties of spray dried powders, as determined by SEM and DSC, were dependent on the ratio of lecithin/calcium caseinate.

A hot melt coating process in a fluidized bed apparatus was used to

coat N-acetylcysteine with a formulation consisting tripalmitin and Polysorbate 65 to achieve masking of unpleasant sour taste in an immediate release product (Stocker et al., 2017). Quality risk management approach was applied to optimize the developed formulation. Efficiency of taste masking (0.4 % of N-acetylcysteine release in dissolution medium from coated composition), particle size (D_{90} = 820 μm) and immediate release profile (85 % within 30 min) were critical quality attributes of the coating composition. The impact of critical material attributes (25–40 % coating amount and 10–20 % emulsifier content) and critical process parameters (air flow rate, atomization air pressure and spray rate) were assessed with statistical design of experiment using a fractional factorial design.

Wax coating through a fluidized bed coating process offered suppression of bitter astringent taste of indeloxazine hydrochloride without compromising bioavailability (Sugao et al., 1998). Sieved particles of indeloxazine hydrochloride were granulated and coated in a fluidized bed granulator with a side spray coating method. Granules of median diameter 130 μm were subjected to coating with different coating solutions containing hydrogenated oil (Lubri wax 103), sucrose fatty acid ester (Ryoto Sugar Ester S-370 F) or stearic acid. Coated particles consisting hydrogenated oil and Pluronic-F68 were subjected to a heat treatment, at a temperature higher than the melting point of Pluronic-F68, in an air stream oven for achieving taste masking and bioequivalence to a commercial tablet.

Table 4gives a summary of some research reports on spray drying/congealing and fluidized based taste masking methods where lipids have been used for masking taste.

2.5.3. Spray congealing

Spray congealing or spray cooling/chilling method is another widely applied technique for masking the bitter taste of drugs. In this process where the molten material is sprayed through nozzles into a chamber containing cool air to produce spherical uniform shaped particles (Cavallari et al., 2005). This process is capable of producing uniform free flowing spherical particles from molten materials and low bulk density products. Melting point of lipid, viscosity/rheology of the formulation and temperature of air are some critical processing parameters. Wheel speed and feed rate control the size range of produced particles (Bertoni et al., 2019; Ilic' et al., 2009; Passerini et al., 2006).

The unpleasant taste of cefuroxime axetil in Zinnat™ suspension is successfully masked with stearic acid while formulating microspheres (Robson et al., 1999). Robson and coworkers prepared stearic acid coated microspheres of cefuroxime axetil with a spray chilling method and assessed for drug release, integrity of microspheres and interaction with media and media components (Robson et al., 1999, 2000a, 2000b). Rapid drug release and loss of physical integrity of microspheres was observed in buffers at pH 7 and pH 8 while less release and intact microspheres were found in buffer at pH 5.9. Interaction of stearic acid with alkaline media was suggested for observing rapid release rate of drug from stearic acid microspheres. These observations facilitated the need of acceptable taste masking in oral cavity and a rapid and complete release in the lower part of intestine. In another study by the same group it was identified that the release of drug from stearic acid depends on both pH and composition of the release media (Robson et al., 2000a). Release studies, coupled with flame emission spectroscopy, and scanning electron microscopy revealed the role of sodium ions in achieving significantly high drug release in the alkaline media. Differential scanning calorimetry (DSC) and high sensitivity DSC also confirmed the role of interaction of microspheres and media composition (Robson et al., 2000b)

Spray chilling method was used to prepare taste masked microspheres of a model drug benzoic acid from stearic acid and palmitic acid (Qi et al., 2008). The melt of fatty acids at 80 °C was mixed with drug and transferred to a spray chiller. Air pressure was used to spray the molten mixture through a nozzle and the solidified microparticles were obtained at room temperature from the collecting chamber. An interaction between alkaline media and palmitic/stearic acid microspheres resulted in formation of acid soap which has a significant impact on drug release. Significant dissolution enhancement may be achieved with the use of mixed fatty acid microspheres due to lower lattice energy of such mixtures and thereby increasing formation of hydrated acid soaps or via releasing fatty acid of low molecular weight into the dissolution/release medium. This study emphasized on a judicious selection of combination of fatty acids as one of the critical formulation parameters.

A molten mixture of glyceryl monostearate, aminoalkyl methacrylate copolymer E and micronized paracetamol was spray congealed to obtain taste masked (Nitanai et al., 2012).A 3^3 Box–Behnken design was used to optimizeloading (%) of paracetamol, Eudragit E and particle diameter for drug release at pH 4 and pH 6.5. A new time-dependent diffusivity model, which assumed the change in the drug's diffusion coefficient, was proposed to predict the drug release from wax particulates. The criterion of 15 % drug release within 10 min at pH 6.5, 90 % of drug release within 30 min at pH 4.0 and a wax matrix particle size of 400 μm was used to predict the wax matrix formulation. With 8.25 % Eudragit E and 15 % paracetamol loading in wax matrix, the experimental drug release profile matched with the predicted release profile.

The effect of stearic acid/cetyl alcohol/cetyl esters and paracetamol on the rheological properties of paraffin wax was evaluated in spray congealed microparticulte formulations of paracetamol (Ouyang et al., 2019, 2018a, 2018b). The newtonian flow of molten paraffin wax was changed to plastic flow with the addition of 20 % paracetamol and pseudoplastic flow with stearic acid/cetyl alcohol/cetyl esters addition at 40 % level. Further, the paraffin-wax blends produced smaller size microparticles via a spray congealed process. Degree of drug coating and drug release was also higher with the addition of lipidic excipients in the wax matrix. Lowest drug release was obtained with wax-cetyl alcohol based microparticles of paracetamol. Cetyl esters and cetyl alcohol were found to have the potential for masking taste in wax matrices on the basis of in vitro drug release (Ouyang et al., 2019).

Octadecanol, glyceryl monostearate and ibuprofen in 6:1:3 ratio were taken together and used to prepare taste masked microparticles via a spray congealing process (Qin et al., 2019a, 2019b). Internal 3D structure and external morphology of the microspheres were studied with synchrotron radiation X-ray micro-computed tomography for finding their possible role in taste masking and drug release. Optimization of formulation on the basis of internal structure and surface characteristics of microparticles was proposed for rational designing of taste masked microspheres.

Clarithromycin wax matrices consisting of clarithromycin, glyceryl monostearate and Eudragit E in 3:6:1 ratio were prepared by a spray congealing agglomeration process for masking the bitter taste of clarithromycin (Yajima et al., 1996). Liquid feed rate and atomizer wheel speed were optimized to achieve smooth surface small spherical matrices capable of masking the bitter taste of clarithromycin.

2.5.4. Melt granulation method

Melt granulation is defined as the method of adding molten binder to a powder mixture for obtaining agglomerates of desired properties (Desai et al., 2013). Apart from binders, active ingredients and other excipients like plasticizers, lubricants, etc.are also added to the powder mixture (Passerini et al., 2010). Lipids like stearic acid, paraffin wax, carnauba wax, glyceryl behenate, hydrogenated castor oil, glyceryl stearate, glyceryl palmitostearate, etc have been used extensively as binders in melt granulation. Melt granulation may be conducted in a fluidized bed apparatus where mixing, heating/cooling and coating are performed in a single apparatus. On the other hand, granules can be prepared in a high-shear mixer, where the impeller speed and temperature of the heating jacket are some critical process parameters. Granules may be as such used or incorporated into other dosage forms like tablets, capsules, etc. Melt granulation in fluidized apparatus eliminate the use of solvent and thereby improve stability of formulations and also

Table 5
Melt granulation methods used for taste masking by lipids in research woks at a glance.

Lipid	Active Ingredient	Other Important Excipients	Method/Technique	Dosage Form/ Delivery System	Significant Outcome	Reference
Beeswax, Compritol®888 ATO, Gelucire® 50/13, Precirol® ATO5, and Geleol® (glyceryl monostearate) and stearyl alcohol	Divalproex sodium		Solvent free hot melt granulation process	Tablets	Box Behnken design was applied for optimizing concentration of lipid binder, impeller speed, and granulation time on the drug release. Compritol® and Beeswax were found superior in stability, bioavailability and taste masking efficiency.	(Khaled et al., 2020)
Stearic acid	sucrose octaacetate	Pullulan, mannitol	Hot melt method for microspheres	Microspheres incorporated into edible films\	Psychophysical studies in adults found embedding of lipid based microspheres and incorporating flavors in ODFs as an efficient mean of masking bitter taste.	(Smutzer et al., 2020)
Geleol pellets 115 (glycerol monostearate, glyceryl stearate)	Ondansetron HCl		Hot melt fusion granulation	Granules	Lipid (Geleol) and hydrophilic agent (Cab-o-sil) based matrix granules were prepared for controlling the release of drug for the purpose of taste masking.	(Kharb et al., 2014)
Cetyl alcohol, stearic acid and glyceryl monostearate	Satranidazole	Eudragit EPO	Melt granulation	Granules	Stearic acid and drug ratio of 2:1 was found suitable for masking the bitter taste of drug. The assessment was based on in vitro drug release and comparing it with bitterness threshold of the drug.	(Pawar and Joshi, 2014)
Hard fat (Witocan® 42/44)	Sodium benzoate		Melt granulation and tableting	Minitablets comprising coated lipid granules/ coated granules	A single step melt granulation process was used for coating hard fat onto drug particles in a high shear mixer. Melt granulation followed by subsequent compression into minitablets offered acceptable taste masking.	(Eckert et al., 2014)
Stearic acid	Quinine HCl	Sucralose, peppermint oil	Modified hot melt method	Microspheres in edible films	Fast dissolving pullulan edible films comprising stearic acid microspheres were prepared for pediatrics and geriatrics.	(Cherian et al., 2018)
Bees wax and cetyl alcohol	Bromhexin hydrochloride, salbutomol sulphate		Extrusion spheronization and hot melt coating	Coated pellets	Cetyl alcohol and beeswax were equally effective in suppression of bitter taste when assessed through taste panel studies.	(Patil et al., 2011)
Glyceryl palmitostearte (Precirol Ato 5))	Actaminophen		Melt granulation	Taste masked granules incorporated into water dispersible tablets	Taste masked acetaminophen immediate release tablet with 90 % drug dissolution within 15 min.	(Duru et al., 2004)

protect environment from toxic vapors of organic solvents. Despite having these advantages, melt granulation suffers disadvantages of degradation of drugs or lipids due to high temperatures (Desai et al., 2013).

Hard fat (Witocan® 42/44) and sodium benzoate granules were prepared by wet granulation and subsequently incorporated into minitablets for masking the bitter taste and offering flexible pediatric dosing (Eckert et al., 2014).A single step melt granulation process was used for coating hard fat onto sodium benzoate particles in a high shear mixer. Melt granulation was followed by subsequent compression of granules of size 315–630 μm into minitablets. Melt granulation alone was not sufficient to mask the taste. However, the combined approach of melt granulation and subsequent preparation of minitablets was successful in decreasing the release of active ingredient.

Satranidazole was melt granulated in a base of cetyl alcohol or stearic acid or glyceryl monostearate with Eudragit EPO (Pawar and Joshi, 2014). The granules were mixed with mannitol, xylitol, aspartame, vanilla flavor, magnesium stearate and other excipients. Stearic acid, Eudragit EPO and drug ratio of 6:3:1 ratio was found suitable for masking the bitter taste of drug in granules. The taste assessment was based on in vitro drug release and compared it with the bitterness threshold of the drug.

A hot melt method was used to encapsulate quinine HCl and sucralose into stearic acid (5.5:1 ratio by weight) microspheres (Cherian et al., 2018). Subsequently stearic acid microspheres accompanied with peppermint oil and encapsulated sucralose as masking agents were incorporated into edible pullulan films. Incorporation of sucralose microspheres with quinine microspheres was successful in suppressing bitter taste and increased acceptance among volunteers when assessed for temporal dominance of sensation. Sweet sensation for long duration persisted in films which contained sucralose microspheres when compared to films devoid of sucralose microspheres.

Table 5 describes in brief various melt granulation methods for masking bitter taste with lipids.

2.5.5. *Microparticles by emulsification*

Emulsions based methods for preparing microparticles have been used widely for masking bitter taste. However, the use of organic solvent and subsequent drying of microparticles is a limitation with all such methods. In one interesting study, slow releasing and taste masked lipid encapsulated microparticles of acacia tannin extract were prepared by solid-in-oil-in-water emulsion method (Adejoro et al., 2018). A slow in vitro release of tannin of about 20 % was observed from microparticles in the acetate buffer after 24 h. Table 6 enlists some literature reports on taste masking with lipids by fabricating particulates using various emulsification processes.

2.5.6. *Drug delivery systems, specific bitter inhibitor and Misc. Methods*

With the advancement of reducing development costs of novel drug delivery systems and advancements in robust preparation methods, liposomes and nanoparticles have figured as the latest means of achieving taste masking in oral pharmaceutical formulations (Table 7).

Table 6
Various emulsification methods for preparing lipid based taste masked particulates.

Lipid	Active Ingredient	Other Important Excipients	Method/Technique	Dosage Form/ Delivery System	Significant Outcome	Reference
Lard and palm oil	Acacia tannin extract	Tween 80	Solid-in-oil-in-water method and melt dispersion method	Microparticles	Slow releasing and taste masked lipid encapsulated microparticles were prepared by solid-in-oil-in-water emulsion method. About 20 % release of tannin from microparticles in acetate buffer after 24 h.	(Adejoro et al., 2018)
Stearic acid	Enrofloxacin	Chitosan, alginate	Solid dispersion (Emulsion solvent diffusion) with staric acid and chitosan-sodium alginate microparticle coating	Microparticles (double coated)	Inner coating of stearic acid and an outer coating of chitosan-sodium alginate improved palatability of enrofloxacin in pigs.	(Liu et al., 2017)
Glyceryl monostearate	Quinine sulphate	Pluronic F68, Pluronic F127, Tween 80	Ultrasonic solvent emulsification	Solid lipid nanoparticles	Optimized formulation release drug to negligible extant in simulated salivary fluid and complete release in simulated gastric fluid. Taste masked solid lipid nanoparticles were presented as suitable dosage form for pediatrics/geriatrics	(Dandagi et al., 2014
Capmul® MCM L (glycerol monocaprylocaprate), Captex® 355 (caprylic/capric triglyceride), Imwitor® 408 (propyleneglycol caprylate), Myglyol® 840 (propylene glycol dicaprylate/dicaprate) and Isopropyl myristate	Phenobarbital	Polyoxyl 40 hydrogenated castor oil (Cremophor RH40), caprylocaproyl polyoxyl-8-glycerides (Labrasol) and Transcutol P (diethylene glycol monoethyl ether)	Emulsification	Microemulsions and self-emulsifying drug delivery systems	Developed dosage forms were optimized for taste masking with electronic tongue.	(Monteagudo et al., 2014)
Precirol® ATO 5	Loratadine	Polyvinyl alcohol	O/W emulsification following by congealing	Micropaticles, suspension	Demonstrated the limited use of solid lipid particles for taste masking in solid dosage forms. Liquid suspensions were not able to suppress bitterness.	(Milak et al., 2006)
Soya bean oil, egg lecithin	Diazepam	Poloxamer, alpha-tocopherol and parabens	Emulsificaton	Submicron emulsions	Developed emulsion formulation were not stable and unsuccessful in masking bitter taste of dizepam.	(Gajewska et al., 2001)
Witepsol® H-37, egg lecithin (Lipoid E80), oleic acid, stearic acid, glyceryl tripalmitate			Emulsification in a high shear homogeniser	Lipospheres	The developed formulation was unable to mask the bitter taste of diazepam (at 0.4 % by weight).	(Sznitowska et al., 2020)

In 1993, lipoprotein suspension (Phosphatidic acid and β lactoglobulin) were reported as a specific inhibitor of bitter taste of quinine HCl, papaverine HCl and isoleucine by measuring electrical thresholds of glossopharyngeal nerve and psychophysical tests in human (Katsuragi and Kurihara, 1993a, 1993b). Psychophysical tests in humans with 12 bitter drugs/substances had strengthened the position of phosphatidic acid and β lactoglobulin as specific bitterness inhibitors (Katsuragi et al., 1995). In a subsequent study by the same research group, phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol and phosphatidic acid inhibited bitterness of quinine in granules which paves the way for use of these specific bitter taste inhibitors in food and drug products (Katsuragi et al., 1996). All these initial studies led to the successful commercialization of a phospholipid cocktail product Benecoat BMI 60 (Kao Corp., Japan) which contained phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol and phosphatidic acid. In one another interesting trial in patients, electrogustometer measurements revealed significant effect of Benecoat BMI 60 in reducing unknown phantom sensations of bitterness, bitterness due to aloe juice effect and constant bitterness sensation in mouth due to fissured tongue and brain infarction (Ishimaru et al., 2001). Flavoured BMI-60 was found best for masking taste of solutions of commercial products, tablets of Polymyxin B sulphate, Sulfamethoxazole and trimethoprim in combination, as assessed through taste panel studies in children and adults (Saito et al., 1999). Benecoat BMI-40 (Kao Corp., Japan) successfully masked bitter taste of paracetamol in chewable tablets without hampering bioavailability (Suzuki et al., 2003).BMI-40 and aspartame/saccharin were used as taste masking agent and sweetener in chewable tablets prepared with Witocan® H and combination of Witocan® H and Witocan® 42/44 (Suzuki et al., 2004). The use of bitter inhibitor and other approaches for masking taste with the use of lipids are provided in Table 8. Table 9 provides some approaches described in patents for masking bitter taste with the use of lipids as functional materials.

2.6. Regulatory and pharmacopoeial status of lipids

Most of the lipids are devoid of serious immunological effects and stability issues (Shrestha et al., 2014). At present most drug regulatory authorities allow the use of lipids in pharmaceutical products. As per US Code of Federal Regulations (21 CFR 172.515), lipids are considered as

Table 7
Use of novel drug delivery systems like liposomes and nanoparticles for masking bitter taste of drugs.

Lipid	Active Ingredient	Other Important Excipients	Method/Technique	Dosage Form/Delivery System	Significant Outcome	Reference
Phytantriol	Cefpodoxime proxetil		Lyotropic liquid crystalline nanoparticles	Lyotropic liquid crystalline nanoparticles	Taste masked lyotropic liquid crystalline nanoparticles were prepared and assessed for taste masking efficiency using LMTS (TS-5000Z)	(Fan et al., 2019)
1,2-Dipalmitoyl-sn-glycero-3-phosphorylglycerol sodium salt, Lipoid EPC, medium changs triglyceride, stearic acid	Atomoxetine HCl	Polyethylene glycol (PEG) 4000	Reversed lipid-based nanoparticle, Hydration of lipid film	Reversed lipid-based nanoparticles	Reversed lipid based nanoparticles were capable to suppressing bitter taste of drug without any burst effect and increased bioavailability of atomoxetine HCl.	(Zhang et al., 2020)
Distearoyl-sn-glycero-3-phosphocholine and cholesterol	Mefloquine		Liposomal encapsulation	Liposomal encapsulation	A high drug-to-lipid ratio of 0.1−0.2 in liposome was found successful in masking bitter taste in a palatability assessment performed using electronic tongue. Developed liquid formulation was lyophilized to powder form which was for more than 3 months and assessed bioavailable (81−86%) in mice.	(Tang et al., 2017)
Phospholipids, cholesterol	Loratadine		Thin film hydration method	Liposome intermediate and suspension intermediates were incorporated in oral dissolving film	Taste masking was achieved by encapsulating drug in liposome but limited drug absorption and bioavailability was a major limitation of liposome oral dissolving film.	(Zhu et al., 2018)

Table 8
Bitter inhibitor and the use of other approaches in masking taste.

Lipid	Active Ingredient	Other Excipients	Method/Technique	Dosage Form/ Delivery System	Significant Outcome	Reference
Egg phosphatidylcholine (Lipoid E80) and medium chain triglycerides	Azithromycin and roxithromycin		High pressure homogenization and subsequent freeze drying	Reverse micelles	Reverse micelles did not hamper drug release/bioavailability while providing acceptable taste masking.	(Huang et al., 2018)
Octadecanol	Paracetamol	Beta cyclodextrin	Lipid coating and cyclodextrin complexation	Lipid microspheres and inclusion complexes	Synergistic bitterness suppression was achieved with the use of lipid microspheres and cylodextrin complexation as assessed by electronic tongue.	(Li et al., 2014)
Glyceryl monostearate	Paracetamol	Aminoalkyl methacrylate copolymer E (Eudragit® EPO) and ethylcellulose	Melt of lipid was mixed with other components and then solidified in a mould.	Wax matrix	Multiple regression analysis, 3^2 full factorial design with two center points, was applied to find optimum ratio of paracetamol: Eudragit® EPO:ethylcellulose: glyceryl monostearate as 30:7:10:53 for an acceptable release of paracetamol for taste masking.	(Shiino et al., 2010)
Gelucire 39/01	Tramadol	Ion exchang resin (Tulsion 335)	Lipids directly compressed with drug-resin complex and other excipients	Drug resin complex, mouth dissolving tablets	Amounts of Gelucire 39/01 amount crospovidone were optimized by a 3^2 factorial design. A linear decrease in disintegration time of tablet with increase in Gelucire 39/01 amount.	(Madgulkar et al., 2009)
Witepsol® H-15, W-35, S-55, E-75 and E-85, and Witocan® H and 42/44		Benecoat BMI-40, cocoa powde, sucrose, xylitol, saccharin, saccharin sodium, aspartame and sucralose	Casting method	Chewable tablets	BMI-40 and aspartame/saccharin were used as taste masking agent and sweetener in chewable tablets prepared with Witocan® H and combination of Witocan® H and Witocan® 42/44.	(Suzuki et al., 2004)
Cacao butter, Witepsol W-35	Enrofloxacin	Carrageenan or sodium alginate	Solid lipid core prepared by melting-cooling method	Coated beads	The lipid core of beads decreased the drug diffusion at ambient temperature and thereby prevented migration of drug to the surface of the beads.	(Kim and Choi, 2004)
Hard fat (Witepsol H-15) and cacao butter	Paracetamol	Sucrose, cocoa powder and a commercial lecithin based taste masking agent (Benecoat BMI-40)	Direct compression	Chewable tablets	Witepsol H-15 with sucrose and with and without Benecoat BMI-40/cocoa powder were found suitable for masking bitter taste without hampering bioavailability.	(Suzuki et al., 2003)

Table 9
Some methods/techniques used for masking taste with lipids reported in patents.

Lipid	Drug/ food / nutraceutical product	Method of taste masking	Critical process/ formulation parameters	Reference
Sleceted from the group of GM, Glyceroldibehenate, Glyceryl behenate etc.	Raltegravir, ritonavir, rilpivirine, tenofovir, tipranavir, etc.	One-stop solution method	Melting point etc.	(Lee et al., 2019)
A hard fat, either glycerol trimyristate or glycerol distearate, and optionally, glycerol dibehenate	Sodium benzoate	Cold solvent-free extrusion	Speed of spheronization plates etc.	(Breitkreutz and Matilainen, 2010)
Lipid or lipid mixture (insoluble in water)	Praziquantel		Particle size etc.	(Kft, 2013)
Selected from the group of GM, glyceryl monopalmitateetc.	Selected from the group of macrolide antibiotics anti-inflammatory etc.	Selected from the group of microencapsulation, spray drying etc.	pH, drug-lipid ratio	(Menjoge and Kulkarni, 2008)
Mixture of phosphatidic acid (PA) and phosphatidylinositol (PI)	Various foods like fruits, vegetables, dairy products etc.		Mixing speed, etc.	(Goto et al., 2007)
Selected from the group of starch; cellulose; xanthan gum etc.	Nutritional composition comprising a nucleoside and/or a nucleotide etc.		Drug-lipid ratio	(De Kort et al., 2016)
Stearic acid	Cefuroxime axetil	Hot melt extrusion technique	Drug-lipid ratio, feed rate, etc.	(Nande and EDATHUMPADIKKAL, 2013)
Wax or a fatty acid glycerol ester alone or mixture	Selected from acetaminophen, guaifenesin, ibuprofen, etc.	Direct compression method and modified fluidized bed coating method	Drug-lipid ratio, etc.	(Szamosi et al., 2007)
Selected from the group of phospholipids, lecithins, ceramides etc.	Selected from the group of guaifenesin, dextromethorphan HBr, phenylpropanolamine etc.	Wet granulation method	Melting temperature of lipids, pH, etc.	(Howard and Purvis, 2017)
Glycerophospholipid component or sphingoglycolipid etc.	Collagen peptide containing any composition	Direct compression method	Drug-lipid ratio	(Kitahara and Okada, 2016)
Wax with non-polymorphic hydrogenated oil etc.	Selected from the group of acetaminophens, ibuprofen, etc.	Hot melt extrusion method	Melting temperature of lipids, etc.	(Reo and Johnson, 1999)
Low- weight polyethylene glycol (PEG)	Pseudoephedrine, acetaminophen or antihistamine, etc.		Molecular weight etc.	(Moros et al., 2003)
Selected from glyceryl tripalmitate and glyceryl tristearate with polysorbate 65	N-acetylcysteine	Hot melt coating method	Temperature, melting point etc.	(Becker et al., 2016a)
Selected from hard fat, like glycerol etc.	Sodium benzoate	Cold solvent-free extrusion method	Temperature of release etc.	(Breitkreutz and Matilainen, 2010)
Fat	Selected from pharmaceutical ingredients or nutraceutical agents etc.		Temperature, concentration of drug, etc.	(Ervin and Ervin, 2014)
Different forms of vitamin E and fat	Menthol		Ratio of active agent etc.	(FIRRELL et al., 2013)
Carnauba wax and magnesium aluminium silicate	Guaifenesin	Melt granulation	Melting point of lipid	(Mozda, 1988)
GM	Cimetidine	Granulation	Speed of paddle, etc.	(Gottwald et al., 1991)
Gelatin and mixture of partially hydrogenated soybean oil and GM	Gabapentin	Coating	Compression force, etc.	(Cherukuri and Chau, 1991)
Stearyl stearate	Selected from acetaminophen, ibuprofen, etc.	Spray congealing method/ Fluid bed process	Melting point of lipid etc.	(Gowan and Bruce, 2003)
Soybean lecithin	Talampicillin hydrochloride	Fluid bed process	pH, spray rate, air pressure, etc.	(Block et al., 1990)

safe and hence accorded the status of GRAS. However, the use of some lipids at high amounts may demonstrate toxicity which suggest their use in safe amounts (U.S. Food and Drug Administration, 2020a, 2016; Kalepu et al., 2013; U.S. Food and Drug Administration, 2004). This safe amount of lipids may vary among dosage forms and may also differ with different routes of administration. Inactive Ingredient Guide (IIG) may be consulted for finding the use of lipids in marketed and approved pharmaceutical formulations. A formulation scientist may take help from IIG for selecting lipids and their acceptable amounts for developing new lipid formulations and eliminate the hurdles of review process (U.S. Food and Drug Administration, 2020b).

Monographs of various lipids have been included in official pharmacopoeias like Indian Pharmacopoeia (IP), U.S. Pharmacopoeia National Formulary (USP NF), Japanese Pharmacopoeia (JP), European Pharmacopoeia (PhEur), British Pharmacopoeia (BP), etc. The pharmacopoeial status along with the USFDA and EMA acceptance status are furnished in Table 10 (Pharmacopoeia Online, 2020; U.S. Food and Drug Administration, 2020a; European Medicines Agency, 2020; The Japanese Pharmacopoeia, 2016; Walsh et al., 2014; Indian Pharmacopoeia, 2010; British Pharmacopoeia, 2009).

3. Taste assessment

3.1. Mechanism of bitter taste perception

The starting of 21st century marks the discovery of taste receptors specific to bitter taste (TAS2Rs or T2Rs), sweetness (T1R2 + T1R3) and umami taste (T1R1 + T1R3) (Wu et al., 2020a, 2020b). However, salt and sour sensation is linked to various ion channels, viz. acid sensing ion channels, polycystic kidney disease 2-like 1 protein, epithelium sodium channels and hyperpolarization activated cyclic nucleotide-gated channels.

The perception of bitter taste is a biochemical process which starts with the initiation of a biochemical response when a chemical interacts with taste receptors located in the taste buds. Nerves conduct this biochemical response to the brain where this stimulus is further propagated leading to cognitive sensation of bitter taste. Bitter taste receptors (TAS2Rs or T2R), initially identified as type II taste receptor

Table 10
Regulatory and pharmacopoeial status of somelipids for their use in pharmaceutical products.

Lipids	Pharmacopoeal status	USFDA acceptance status	EMA acceptance status
Glyceryl monostearate	Present in IP, BP, USP NF*, JP	Accepted	Accepted
Lecithin	Present in IP, JP, USP NF*	Accepted	Accepted
Carnauba wax	Present in JP, IP, USP NF*	Accepted	Accepted
Bees wax	Present in JP, USP NF*, PhEur,	Not listed	Accepted
Glyceryl distearate	Present in USP NF*, PhEur	Not listed	Not listed
Glyceryl dibehenate	Present in USP NF*, PhEur	Accepted	Not listed
Glyceryl tristearate	Not listed	Accepted	Not listed
Glyceryl trymyristate	Not listed	Not listed	Not listed
Glyceryl palmitostearte	Not listed	Not listed	Not listed
Stearic acid	Presentin JP, USP NF*, IP, BP	Accepted	Not listed
Tripalmitin	Not listed	Not listed	Not listed
Paraffin wax	Present in JP, IP, BP	Accepted as liquid form	Accepted as liquid form
Hydrogenated palm oil	Present in USP NF*	Accepted	Not listed
Cetyl alcohol	Present in USP NF*, IP, BP	Accepted	Not listed
Palmitic acid	Present in JP, USP NF*, IP, BP	Accepted	Accepted
Hydrogenated phosphatidylcholine	Present in JP as phosphatidylcholine	Accepted as phosphatidylcholine	Accepted as phosphatidylcholine
Phospholipids	Present in USP NF* (egg Phospholipids), JP, IP, BP	Accepted (egg Phospholipids)	Accepted (egg yolk Phospholipids)
Phosphatidic acid	Not listed	Not listed	Accepted as phosphatidylglycerol
Polyethylene glycol (PEG)	Present in IP, JP, USP NF*, BP, PhEur	Accepted	Accepted
Soya-bean oil	Present in JP, BP	Not listed	Accepted
Hydrogenated coco-glycerides	Present in USP NF*, JP, PhEur	Not listed	Not listed
Lauroylpolyoxylglycerides	Present in PhEur, USP NF*	Accepted	Not listed

[*USP 32 NF 27 used here].

cells, are seven-transmembrane G protein–coupled receptors located in taste buds of the tongue(Lu et al., 2017; Max and Meyerhof, 2008).The signal transduction pathway of bitter taste receptors (TAS2Rs) shares gustducin, a heterotrimeric G protein as a common signaling molecule, with sweet and umami receptors (Lu et al., 2017). Upon interaction of T2R with bitter taste molecules, gustducin dissociates into its subunits, viz. Gnat 3, Gβ3, and Gγ13. Subsequently Gy13 subunit activates phospholipase C (PLCβ2) which in turn stimulates inositol trisphosphate receptor (IP3R) for the release of Ca^{2+} leading to an influx of Na + through TRPM5 channels. Sodium ion influx causes depolarization of the cells and subsequent ATP release via CALHM1 or gap junction hemichannels. Purinergic receptors on nerves are activated with ATP allowing the conduct and propagation of impulse to the central nervous system for sensing bitter taste.

3.2. Lipid membrane taste sensors (LMTS)

Potentiometry (Ciosek and Wr, 2006; Szpakowska and Marja, 2009; Toko, 1998) based taste sensors have established themselves in taste assessment of pharmaceutical and nutraceutical products. Potentiometric sensors use either several ion selective electrodes (ISE) or all-solid- state electrodes (ASSE). Both of these electrodes comprise taste sensing films made from lipids, poly vinyl chloride (PVC) and plasticizer. ISE contains KCl as the electrolyte solution. While in ASSE electrically active polymer, attached to a glassy carbon disc, is used as an electron transducer. Upon immersion of a lipid-polymer membrane in aqueous solution, acidic groups of lipid molecules dissociate with subsequent formation of an electrical double layer leading to a finite membrane potential which depends on the composition of aqueous solution. Thereby, lipid-polymer membranes act as taste sensing membranes which convert taste stimuli to electric signals. Lipid-polymer membranes fitted into electrodes coupled with a reference electrode have advanced themselves as Lipid Membrane Taste Sensors (LMTS) which serve as a great tool in qualitative and quantitative assessment of taste and palatability of pharmaceutical and food products.

LMTS use lipids, polymers and plasticizers in various compositions for preparing sensing membranes of varying electrical charges and hydrophobicity. Lipids upon ionization in solution exhibit hydrophilicity and tranduce electrical signals in a non-conducting polymer like Polyvinyl Chloride (PVC). Polymer encapsulates and immobilizes the lipid. LMTS rely heavily on the properties of lipids and plasticizers for hydrophobic and electrical properties of sensing membranes (Anand et al., 2007). Lipids and plasticizers of long carbon chain length confer stability and hydrophobicity to sensing membranes. Addition of plasticizers improves flexibility of membranes. Generally, a bitterness sensor requires membrane of high hydrophobicity and a positive charge. The sensor for sour taste possesses a neutral charge membrane with high hydrophobicity. Upon interaction of sensing membranes with taste substances, the electric charge density of sensing membrane changes resulting in membrane potential changes which are measured with a potentiometer (Wu et al., 2020a, 2020b). Lipid-polymer membranes are optimized for electric charge density and hydrophobicity to achieve selectivity in response to each taste stimuli. Hence, the output of the taste sensor differs with different taste while similar tastes generate similar patterns.

Taste sensing lipid membranes are generally prepared by deposition of Langmuir-Blodgett film or dip-coating or a simple solvent evaporation method in a Petri plate. Prepared membrane is cut to a suitable size and attached to an electrode, generally an Ag/AgCl electrode which is filled with a saturated solution of KCl. This sensor electrode is preconditioned for 1 day by immersing in a standard solution. This preconditioning allows lipid molecules to reach at interface due to the presence of hydrophilic groups in lipids. Such different electrodes are coupled together with a reference Ag/AgCl electrode to form multichannel taste sensing systems or electronic tongues (Toko et al., 1994a, 1994b; Wu et al., 2020a, 2020b). LMTS are built to measure a relative value (V_s-V_r) and "change in membrane potential caused by adsorption" (CPA) for test solutions. A time of 30 s is generally considered sufficient for measuring membrane potential and four measurements for each sample are recommended (Wu et al., 2020a, 2020b). After each measurement the electrodes are cleaned with a cleaning solution.

CPA value corresponds to the ability of substances to be adsorbed onto taste receptors of the human tongue. Hence, CPA values are taken as predicted bitterness intensities are correlated to bitterness intensity scores of human gustatory sensation tests (Uchida, 2014). Sensor data (CPA values) may be further processed via Principal Component Analysis (PCA) or Principle Least Squares (PLS) regression analysis for developing mathematical models for predicting taste (Immohr et al., 2017; Li et al., 2020). These developed models are subsequently validated for prediction of taste scores (Nakamura et al., 2015).Euclidean distance refers to distance between test and a control substance on a PCA map. Euclidean distances have been employed for comparing taste and

palatability of pharmaceutical formulations (Chay et al., 2018; Ito et al., 2013; Kataoka et al., 2008; Eckert et al., 2014; Petrovich et al., 2016). Large Euclidean distances from control taste indicate less intensity while shorter Euclidean distances indicate strong intensity of taste. Equal Euclidean distance indicates same intensity of control and test samples. Similarly, similarity/dissimilarity of Euclidean distance helps in discriminating formulations on the basis of taste/palatability.

Toko and coworkers invented the first potentiometric taste sensor based on lipid-polymer membranes as transducers of taste (Toko et al., 1994a, 1994b; 1998). This lipid-polymer membrane taste sensor equipped with a multichannel electrode was capable of detecting electric potential patterns from taste substances in a drastically similar way as the human tongue. Eight different lipid analogues dioctyl phosphate (DOP), cholesterol, oleic acid, decyl alcohol, trioctyl methyl ammonium chloride (TOMA), oleyl amine, distearyl dimethyl ammonium bromide and trimethyl stearyl ammonimum chloride with polyvinyl chloride (PVC) and dioctyl phenylphosphonate (DOPP) as plasticizer were used to prepare films of 160/200 μm to be fitted to a Ag/AgCl multichannel electrode (Toko et al., 1994a, 1994b; Iiyama et al., 1994). These lipid analogues were selected for their similarity of functional groups of lipids which are present in the biological membranes. Mixtures of lipids, DOP: TOMA in ratio 5:5, 3:7 and 9:1, have also been optimized for fabricating taste sensing lipid-PVC membranes specific to tastes (Toko, 1998).Lipid membranes detect surface potential at the interface of lipid membrane and solution of the taste substance and transmit it to Ag wire of the multichannel Ag/AgCl electrode. Voltage difference between a reference electrode and multichannel taste sensor is measured. In this sense, these lipids membranes were also referred to as taste sensing membranes. Taste sensor was designed for quantitative and qualitative analysis of five tastes, viz. bitterness, sourness, saltiness, sweetness and astringent (Toko et al., 1994a, 1994b; Iiyama et al., 1994). This LMTS was conceptualized for global selectivity of assessing taste interactions of substances ranging from simple solutions to complex multicomponent formulations. Global selectivity of LMTS refers to the ability to respond to different taste substances at the same time and a consistent response to each taste substance every time.

Okahata and coworkers successfully tested synthetic lipid membranes for detection of odorous (p-ionone, skatol, heptanoic acid, octanol, coumarin, and amyl acetate) or bitter (brucine, strychnine, quinine, picric acid, octaacetylsucrose, and naringin) substances (Okahata et al., 1990). Synthetic multibilayer-immobilized lipid film ($2C_{18}N^{+}2C_1/PSS^{-}$) and a cast film made of phospholipids were coated onto a quartz-crystal microbalance. Electrical responses due to adsorption of bitter/odor substances were measured as membrane potential/-membrane resistance changes and advocated as useful means for chemoreception in biological membranes.

A sensor specific to bitter taste assessment was improved by optimizing composition of a lipid phosphoric acid di-*n*-decyl ester (33 %) and two plasticizers, viz. tributyl o-acetylcitrate (17 %) and bis(1-butylpentyl) adipate (100 %) within sensor membranes (Wu et al., 2017). This new optimized composition was found more durable when compared to commercial bitterness sensor (BT0) whose response decreases with time. This effect is so pronounced that the commercial sensor (BT0) is of no use after two years. Liquid chromatography-tandem mass spectrometry and gas chromatography-mass spectrometry (GC–MS) revealed hydrolysis of tributyl o-acetylcitrate and thereby generation of negative charge substances for increasing surface charge density with time. A low ratio of tributyl o-acetylcitrate was used to suppress hydrolysis and improve sensitivity.

Kumar and coworkers fabricated a sweet sensitive gold nanoparticle modified artificial lipid membrane for analysis of sweet taste (Kumar et al., 2018). The lipid-polymer membrane composed tetra-dodecylammonium bromide (TDAB) as lipid, DOPP as plasticizer and polyvinyl chloride (PVC) as polymer. This sensing membrane was modified by incorporating citrate coated gold nanoparticles which affected fluidity, hydrophobicity and surface charge density. A higher sensitivity to fructose molecule was achieved with modified sensing membrane.

Sharma and coworkers developed a taste sensor utilizing artificial lipids onto gold/carbon printed electrodes (Sharma et al., 2015). Methyl trioctyl ammonium chloride as lipid and DOPP as plasticizer were used to prepare membrane for bitterness sensor. TDAB as lipid and DOPP as plasticizer were used for fabricating membrane for sweet taste sensor. Three lipids, viz. tetra dodecyl ammonium chloride (TDAC), 1-tetra decanol (DC) and DOPP, were used to prepare membrane for salt taste sensor. TOMA, oleic acid (OA), bis (2-ethyl hexyl) phosphate and DOPP were used to prepare membrane for sour taste assessment. TDAB, TDAC, TOMA and DOPP were used to prepare lipid membrane for bitter taste evaluation. Lipid(s), plasticizers and PVC in ratio of 1:3:2 were dissolved in organic solvent to obtain a solution which was subsequently evaporated in a flat glass plate to obtain lipid films. Screen printed electrodes and gold electrodes were coated with lipid membranes by a dip drop method. Developed sensors were characterized for their accuracy and reproducibility.

A miniaturized taste sensor chip for its use in a portable taste sensor was developed with a titanium/silver (Ti/Ag) patterning electrode on a polycarbonate sheet (Tahara et al., 2011). As an electrolyte layer, a poly (hydroxyethyl methacrylate) hydrogel (pHEMA) was polymerized and placed onto an Ag/AgCl electrode with a double-faced adhesive tape. Electrolyte layer, pHEMA with KCl, and deposited artificial lipid membrane were used as taste sensing sites. Electrolyte layer and artificial lipid membrane solutions were dropped through holes in a partition. Artificial lipid membrane solution was deposited onto the pHEMA layer and subsequently the sensor chip was conditioned in a solution of KCl and tartaric acid for one day. Five different artificial lipid membrane solutions were used to develop taste sensors for bitterness, saltiness, sourness, umami and astringency. A single artificial lipid TDAC and DOPP and 2-nitrophenyl octyl ether (NPOE) were used as plasticizer, respectively, for developing taste sensing membrane for bitterness and taste sensor for astringency, respectively. Two artificial lipids phosphoric acid di(2-ethylhexyl) ester and trioctylmethylammonium chloride and one plasticizer DOPP were used for fabricating taste sensing membrane for umami taste. A ternary mixture of lipids, viz. phosphoric acid di(2-ethylhexyl) ester, oleic acid and trioctylmethylammonium chloride, was used with DOPP as plasticizer for developing taste sensing membrane for sour taste. Ag/AgCl ink was applied onto a developed chip as a reference electrode. The taste sensing membrane for saltiness contained TDAC and n-tetradecyl alcohol as lipids and DOPP as a plasticizer. Fabricated sensor chipswere used to assess taste of tartaric acid (sour), KCl (salt), monosodium glutamate (umami), tannic acid (astringent) and iso-alpha acid (bitter).Good reproducibility and high accuracy was observed for CPA measurements from fabricated sensor chips when connected to TS-5000Z taste sensing system and upon comparison with the conventional taste sensors of TS-5000Z. However, the reference electrode was considered less stable while measuring electrode potential. To address this issue a reference electrode consisting of pHEMA layer with KCl and a polyvinyl chloride (PVC) membrane was developed (Tahara et al., 2013). PVC-pHEMA layered reference electrode was durable and found stable for potential measurements without interacting with the basic taste samples. The developed sensor chip of size of a USB memory stick was tested fine for in-field evaluations of astringency.

Szpakowska and coworkers developed another LMTS based five channel sensor comprising five ISE and an Ag/AgCl reference electrode and demonstrated its successful use in taste assessment of soft drinks (Szpakowska et al., 2006, 2004). Benzyl hexadecyldimethylammonium chloride monohydrate, elaidic acid, hexadecylamine, cholesterol and 1-dodecanol were used as lipids in preparing taste sensing lipid membranes. Developed taste sensor was found sensitive to sourness, saltiness, bitterness but insensitive to sweetness (Szpakowska et al., 2006). Potentiometric taste sensor with four ASSEs each containing two lipids

Table 11
The use of LMTS taste sensors in assessment of taste and palatability of pharmaceutical/food formulations.

Active Ingredient/Taste Substance	Formulation	Taste Masking Agent	Taste Assessed	Taste Sensor and Equipment/Machine	Remarks	Ref.
Acesulfame K	Solutions	None	Sweet taste	Prototype sweetness and saltiness taste sensors with TS-5000Z	A quantitative method, using nonlinear regression on data obtained from sensors, was developed for sensing taste of acesulfame potassium, a negatively charged artificial sweetener.	(Liu et al., 2020)
Theobromine, theophylline and caffeine	Solutions	None	Bitter taste	Prototype taste sensor with TS-5000Z	The surface of bitter taste sensing membrane (TDAB-PVC-DOPP) was modified with 2,6-dihydroxybenzoic acid or 2,4,6-trihydroxy-benzoic acids for increasing the response of for assessing non-charged bitter substances using LMTS.	(Yoshimatsu et al., 2020)
Atomoxetine hydrochloride	Inclusion complexes	2-hydroxypropyl-β-cyclodextrin	Bitter taste and bitterness suppression	Three bitterness sensor (BT0, AN0, C00) and one astringent sensor (AE1) with SA402B	PCA map was used to assess bitterness suppression.	(Li et al., 2020)
Isoniazid, rifampicin, pyrazinamide, and ethambutol dihydrochloride	Solutions	None	Bitter taste	Three bitterness sensor (BT0, AN0, C00) and one astringent sensor (AE1) of TS-5000Z	Rat brief-access taste aversion (BATA) model and human taste panel studies were compared with TS-5000Z taste sensor assessment of bitter taste of antitubercular drugs.	(Keating et al., 2020)
Diphenhydramine HCl	Solutions	Dipeptides (Asp-Asp and Glu-Glu), and amino acids aspartic acid and glutamic acid	Bitterness and bitterness suppression	AN0 sensor with SA402B	Taste sensor responses and human gustatory sensation tests were correlated and used to investigate binding of bitter drug and taste masking agents with bitter taste receptor (hTAS2R14)	(Okuno et al., 2020)
Diphenhydramine, chlorpheniramine and 45 other bitter drugs from WHO model list of essential medicines for children (2017 ed.)	Solutions	None	Bitterness	AC0, AN0, BT0, AE1 and C00 sensors with SA402B	Drugs having dose number ($D_{0)}$ greater than 1 and fractional bioavailability ($F_{a)}$ lesser than 0.85 correlated well to responses of bitter taste sensors.	(Haraguchi et al., 2019)
Amlodipine besylate in combination with alacepril, benazepril, hydrochlorothiazide, telmisartan and valsartan	Fixed dose combinatons of drugs	Drug-drug interaction	Bittere taste and bitterness suppresssion	AN0 Sensor with SA402B	Electrostatic interaction of negatively charged tetrazole group on valsartan and amino group on amlodipine besylate led to suppression of bitterness.	(Kojima et al., 2019)
Zopiclone and eszopiclone	Tablet dispersed in beverages	Mugicha, sports beverage, lactic acid drink, orange juice and a diluted simple syrup (an 8.5 % sucrose solution)	Bitter taste and bitterness suppression	AC0 sensor with SA402B	Electrostatic interaction of citric acid and drugs was investigated for bitterness suppression.	(Yoshida et al., 2019)
Ranitidine HCl	Inclusion complexes	(2-hydroxypropyl)-beta-cyclodextrin	Bitter taste and bitterness suppression	Three bitterness sensors (SB2AC0, SB2AN0 and SB2C00) and one astringency sensor (SB2AE1) with TS-5000Z	Bitterness suppression was assessed by the Euclidean distance between developed formulations and the pure drug on a PCA score plot. NMR spectroscopy and docking studies revealed mechanistic explanation to complexation and taste masking.	(Chay et al., 2018)
Dextromethorphandi, haloperidol, phenidol, diphenhydramine, papaverine and other bitter compounds (no = 22)	Solutions	None	Bitterness	BT0, AC0, AN0, C00 and AE1 sensors with SA402B	Taste sensor BT0 was found most sensitive to bitterness and its responses of BT0 and AN0 correlated well with the responses of human bitter taste receptor	(Haraguchi et al., 2018)

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Table 11 (*continued*)

Active Ingredient/Taste Substance	Formulation	Taste Masking Agent	Taste Assessed	Taste Sensor and Equipment/Machine	Remarks	Ref.
					hTAS2R14. CPA values of sensor were correlated to bitterness intensity of solutions as perceived by human taste panel. Physicochemical parameters from bitter database were also correlated to taste sensor responses.	
Praziquantel	Various solutions of praziquatel with maltodextrin and cyclodextrins	Maltodextrin, hydroxypropyl-beta-cyclodextrin and sulfobutyl ether-beta-cyclodextrin	Bitter taste and bitterness suppression	TS-5000Z and a self developed sensor comprising artificial lipid trioctylmethyl ammonium chloride/ tetra-dodecyl ammonium bromide/ bis(2-ethylhexyl) phosphate.	Rodent brief-access taste aversion and IC 50 value (concentration of drug inhibiting 50 % of water lick numbers) was more efficient taste assessment methodology for praziquantel and its formulations. An equal taste masking in vivo efficiency of maltodextrin and cylodextrins was reported.	(Münster et al., 2017b)
Quinine HCl	Solutions of quinine HCl	None	Bitter taste	20 different LMTS including sensing membrane of BT0 sensor with TS-5000Z	Taste sensing membranes were optimized for selectivity and durability of bitterness sensor.	(Wu et al., 2017)
Quercetin	Microcapsules	Carnauba wax, shellac and zein	Bitterness and bitterness suppression	C00 sensor withTS-5000Z	Taste sensor was used to assess bitterness suppression effect of different taste masking agents in microcapsules prepared by hot melt extrusion.	(Khor et al., 2017)
Diphenhydramine hydrocholoride		Chlorogenic acid	Bitterness and bitterness suppression	AN0 sensor of SA402B	Bitterness suppression of chlorogenic acid was assessed with taste sensor and NMR studies revealed electrostatic interaction between drug and chlorogenic acid	(Shiraishi et al., 2017)
Donepezil HCl, quinine HCl	Marketed ODFs and ODTs and developed ODFs	Benecoat BMI-40	Bitter taste	BT0 sensor with SA402B	A method for continous mesurement of bitterness for upto 100 s was developed.	(Takeuchi et al., 2017)
Metamizole sodium, pseudoephedrine sulphate	Physical mixtures of drug and coating agents	Hypromellose, carmellose, Eudragit E	Bitterness suppression	Eight different membrane compositions with 16 ISEs	Electronic image sensing of particles in dissolution tests of physical mixture for assessing dissolution and taste masking.	(Wesoly et al., 2017a)
Cetirizine dihydrochloride	Microspheres	Kollicoat SmartSeal (SS) and Kollicoat Protect (KP)	Bitterness suppression	Eight different membrane compositions with 16 ISEs	Chemical image assessment of dissolution samples with electronic tongue for measuring extent of dissolution and taste masking.	(Wesoły et al., 2017b)
Valsartan	Minitablets	Eudragit E	Bitter taste and bitterness suppression	Eight different membrane compositions with 16 ISEs	LMTS with cross selective electrodes were employed for assessing taste masking efficiency of coating and investigating drug release kinetics.	(Wesoly et al., 2016)
Ambroxol HCl, cetirizine dihydrochloride and sodium picosulfate	Cetirizin AL, Ambroxol-ratiopharm® and Laxoberal®	8 Different beverages, viz. tap water, apple juice, fennel tea, fruit tea, carrot juice, cocoa, milk, and Alete meal to drink	Bitterness and bitterness suppression	7 self-developed sensors (TOMA/bis(2-ethylhexyl) phosphate as lipids) and 15 commercial sensors with SA402B	Mixing of medicines with different beverages was assessed with taste sensor for taste masking efficiency. Taste sensor data was correlated with physico-chemical properties of different beverages for demonstrating suitability in bitterness suppression.	(Immohr et al., 2016)
	Solutions	Theasinensin A		BT0 sensor with TS-5000Z	NMR studies and quantum mechanical calculations	(Guo et al., 2016)

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Table 11 (*continued*)

Active Ingredient/Taste Substance	Formulation	Taste Masking Agent	Taste Assessed	Taste Sensor and Equipment/Machine	Remarks	Ref.
Trp-Leu, Leu-Trp, Trp-Met, Trp-Ala, Trp-Tyr, and Trp-Asp			Bitter taste and bitterness suppression		confirmed formation of a complex of theasinensin A and Trp-containing dipeptides resulting in masking of the bitter taste.	
Metformin HCl	Pellets	Hard fat (Witocan® 42/44), glyceryl distearate (Precirol® ATO 5) and glyceryl trimyristate (Dynasan® 114)	Bitterness and bitterness suppression	SB2AAE, SB2CT0, SB2CA0, SB2C00, SB2AE1, SB2AC0 and SB2AN0 sensors with TS5000Z	Euclidean distances on PCA map for drug release samples of 30 and 60 s were used to assess bitterness and its suppression in formulated lipid pellets.	(Petrovich et al., 2016)
Quinine, glucose, glutamic acid, hydrogen chloride, sodium chloride	Solutions	None	Bitter, sweet, salt and sour taste	In house developed lipid membrane taste sensors	Lipid, plasticizers and PVC in ratio of 1:3:2 were used to develop lipid membranes which were deposited onto carbon printed/gold electrodes for developing different taste sensors.	(Sharma et al., 2015)
Caffeine citrate	Solutions	Citric acid, saccharin sodium, maltodextrin	Bitter taste and bitterness suppression	TS5000-Z (8 sensors), αAstree and four other laboratory prototyes of electronic tongues	Comparison on sensor data sets was performed with PCA, Tucker's coefficients and matrix correlation coefficients. More sensors in electronic tongues improve PCA plots with more distinct clustering leading to better discriminating ability of sensors among different formulations.	(Pein et al., 2015)
Cetirizine HCl	Freeze dried (Zydis®) oral formulations	Cyclodextrins (beta and hydroxypropyl beta), ion exchange resins, cherry/ sucralose flavor	Bitter taste and bitterness suppression	Seven sensors with TS-5000Z	Taste sensing system was used to optimize taste masked formulations followed by confirmation of acceptability through a taste panel study.	(Preis et al., 2015)
Aspirin, diclofenac sodium, loxoprofen sodium, indometacin, etodolac, flurbiprofen, naproxen, meloxicam and piroxicam	Solutions	None	Bitterness and sourness	C00 and CA0 sensors with SA402B	C00 sensor was used for accurate assessment of bitterness of acidic drugs in mixed solutions containing sour taste substances and correlated with gustatory sensation tests in humans.	(Yoshida et al., 2014)
Dimenhydrinate	Tablets and films	HPMC (Pharmacoat® 606)	Bitterness and bitterness suppression	SB2AC0, SB2AN0, SB2AAE, SB2CT0, SB2CA0, SB2C00, SB2AE1 sensors of TS5000Z	PCA Score, euclidean distance and loading scatter plots were used to compare performance of taste sensing systems among different laboratories.	(Pein et al., 2014)
Sodium benzoate	Melt graulated formulations and minitablets	Hard fat (Witocan® 42/44)	Bitterness and bitterness suppression	SB2AAE, SB2CT0, SB2CA0, SB2C00, SB2AE1, SB2AN0, SB2AC0, SB2BT0 sensors with TS5000Z	Euclidean distance in PCA map was used to assess bitterness of drug release samples of 10 s and 30 s of different granules and minitablets	(Eckert et al., 2014)
Aspartame	Solutions	None	Sweet taste	A new prototype sensor with TS-5000Z	A taste sensor specific to sweetness of aspartame was developed	(Yasuura et al., 2014)
Iso-alpha acid	Solutions	None	Bitter aftertaste	Several sensors with TS-5000Z	Preconditioning of lipid-polymer membrane with monosodium glutamate and resulting changes in the surface of lipid-polymer membranes was demonstrated for assessing aftertaste.	(Yatabe et al., 2015)
iso-α-acid	Solutions	None	Bitter taste	Sensor having TOMA-DOPP-PVC sensing membrane with SA402B	Adsorption onto lipid membrane and resulting change in electric membrane potential was proposed as a method for assessing bitter taste. The	(Toko et al., 2014)

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Table 11 (*continued*)

Active Ingredient/Taste Substance	Formulation	Taste Masking Agent	Taste Assessed	Taste Sensor and Equipment/Machine	Remarks	Ref.
					hydrophobicity and surface charge density of the membrane affects adsorption of bitter substances on lipid membranes.	
Solifenacin succinate	Vesicare conventional tablets and ODTs	Salting out taste masking system	Bitter taste and bitterness suppression	AC0 sensor with SA402B and α Astree	Gently crushed ODTs were assessed less bitter than heavily crushed ODTs. Hence, ODTs were suggested to be crushed gently for dosage adjustment in pediatrics and to cause least damage to coated particles.	(Haraguchi et al., 2013)
Diclofenac acid form, sodium and potassium salts of diclofenac	Granules, coated granules and ODTs	Eudragit® E PO coating on granules, granulation with Eudragit® E PO and effect of salt form on taste masking	Bitter taste and bitterness suppression	All seven sensors with TS-5000Z	Taste sensing system was capable of identifying and distinguishing diclofenac acid from its salts on the basis of taste modalities. Insufficient discrimination among different formulations was observed with in vitro dissolution studies.	(Guhmann et al., 2012)
Ibuprofen, ibuprofen lysinate, acetaminophen, caffeine, caffeine citrate, sodium saccharin and quinine HCl	Solutions of varying concentration of bitter drugs	Maltodextrin	Bitter taste and bitterness suppression	All seven sensors with TS-5000Z and Astree 2	TS-5000Z was found more reliable and precise over Astree 2. Bitterness suppression by complexation of maltodextrin with ibuprofen and quinine hydrochloride was assessed.	(Woertz et al., 2011a)
Ibuprofen	A generic and several marketed ibuprofen suspension formulations available in German market	Generic suspension contained sweetener (sodium saccharin) and flavor (orange flavor)	Bitterness (& aftertaste), sweetness, sourness, saltiness, umami (& aftertaste), and astringency (& aftertaste)	All seven sensors with TS-5000Z	A rational break down approach for developing taste masked formulations was proposed.	(Woertz et al., 2011b)
Diclofenac sodium	Fast dissolving films	Maltodextrin, sorbitan oleate, mint flavor, licorice flavor, sucralose	Bitterness and bitterness suppression	C00 sensor and AC0 and AN0 sensors of SA402B	While assessing palatability of fast dissolving films, taste sensors were capable of discriminating different formulations.	(Cilurzo et al., 2011)
iso-alpha acid, KCl, tartaric acid, tannic acid, monosodium glutamate	Solutions	None	Bitter, umami, salt and sour taste	Portable taste sensor with several lipid-polymer membranes	Artificial lipid membranes were used as multiple taste sensing sites in a poly (hydroxyethyl methacrylate) hydrogel and KCl medium as electrolyte.	(Tahara et al., 2011)
Diphenhydramine hydrochloride, cetirizine dihydrochloride, dl-chlorpheniramine maleate, epinastine hydrochloride and hydroxyzine dihydrochloride	Solutions	Complexation with cyclodexrins (α, β, γ, and HP-β)	Bitterness and bitterness suppression	Five sensors with SA402	Inclusion complexes were capable of bitterness suppression as assessed by taste sensor and taste panel studies.	(Ono et al., 2011)
Quinine HCl, acetaminophen, sodium saccharin, tartaric acid, KCl	Solutions of quinine HCl	None	bitterness, saltiness, sourness, umami and astringency	Seven sensors with SA402B	ICHQ2 was applied for validating taste assessment method.	(Woertz et al., 2010)
Roxithromycin and Ibuprofen	Microcapsules	Eudragit L30D-55, hypromellose	Bitterness and bitterness suppression	16 sensors (2 electrodes of each type), Among these one electrode with anion selective lipid membrane; Potentiometric multiplexer (EMF 16 Interface)	Chemical image analysis for assessing taste masking potential.	(Janczyk et al., 2010)

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Table 11 (*continued*)

Active Ingredient/Taste Substance	Formulation	Taste Masking Agent	Taste Assessed	Taste Sensor and Equipment/Machine	Remarks	Ref.
Bitter medicinal plants swertiae herb, gentian, bitter orange peel, picrasma wood, phellodendron bark and coptis rhizoma; Astringent medicineal plants Geranii Herb, rhubarb, houttuynia herb, chinese nutgalls and lupli strobilus.	Aqueous extracts of bitter medicinal plants and the Chinese medicines Orengedokuto (3 products), Akadama abdobt (3 products) and Unseiin (4 products)	None	Bitterness and astringency	AC0, AN0, BT0, C00, AE1 taste sensors with SA402	Euclidean distances in PCA map from the control bitter (berberine) and control astringent (tannin) were used to classify bitter and astringent medicinal plants into different groups.	(Kataoka et al., 2008)
Famotidine and quinine sulfate	ODTs	Sucrose or sugar alcohols (sorbitol, erythritol, xylitol, mannitol, lactitol and maltitol)	Bitterness and bitterness suppression	Sweetness (Channel 8) and bitterness sensors (Channel 3) of SA402B	Bitterness of dissolution samples of ODTs were assessed with sweetness and bitterness responsive sensors. Sugar alcohols were found effective for bitterness suppression in initial phase (30 s) of disintegration of studied ODTs.	(Hashimoto et al., 2007)
Claithromycin or azithromycin	Dry syrups	Jellies	Sweetness, bitterness and bitterness suppression	Chennel 3 (bitterness sensor) of SA402B	Bitterness and bitterness suppression was assessed using taste sensor and correlated to human taste panel studies. Chocolate jelly with neutral pH was judged superior in masking bitter taste.	(Tsuji et al., 2006)
Quinine HCl	Solutions	Phospholipid formulation (BMI-60)	Bitterness and bitterness suppression	Decyl alcohol (DA) membrane (DA-DOPP-PVC) with SA402B.	Taste sesnor detected and quantified taste masking effect of BMI-60. Hydrophobic bonding between quinine HCl and phospholipid was confirmed with AFM.	(Shimakawa et al., 2004)
L-isoleucine (L-Ile), Lleucine (L-Leu), and L-valine (L-Val)	Amino acid formulation (Aminoleban® EN)	Flavours and crystal size of amino acids	Bitterness, sweetness and sourness and bitterness suppression	Bitter, sweet and sour sensors with SA402B	Taste sensor and taste panel studies were used to assess taste masking with changes in crystal sizes of amino acids and the effect of various flavours.	(Miyanaga et al., 2004)
Quinine HCl, L-isoleucine and L-phenylalanine	Solutions	Sucrose, aspartame), NaCl, L-aspartic acid, L-glutamic acid, L-histidine, L-lysine and L-arginine, tannic acid and phosphatidic acid (BMI® 40)	Bitterness and bitterness suppression	Channel 3 (bitterness sensor) with SA402B	Taste sensor failed to predict the bitterness-suppression of L-argrinine in quinine HCl and amino acid solutions.	(Ogawa et al., 2004)
Ampicilling, amoxicillin, cefaclor, cefdinir, cefcapene, cefteram, feropenam, cefditoren, azithromycin, erythromycin, carithromycin, midecamycin, fosfomycin, norfloxacin, aciclovir, oseltamivir and amantadine	Tablets, granules and dry syrups diluted with an acidic sport drink Pocari Sweat®		Bitter taste and bitterness enhancement	Channels 1, 2, 3, 8 with SA402B	Taste sensor was demonstrated for successful use in assessment of bitter taste and results correlated with gustatory sensation tests. Acidic sports drink enhanced bitterness of drugs.	(Ishizaka et al., 2004)
Sucrose, fructose, erythritol, glucose, xylitol, sorbitol	Solutions	None	Sweet taste	Sweetness sensor, comprising taste sensing membrane made with two lipids (TDAB and PADE) with DOPP as plasticizer and PVC as polymer, with SA402B	Taste sensor was used to assess sweetness of alternative sweeteners which may replace sugar in food additives.	(Habara et al., 2004)
Clarithromycin	Marketed dry syrup	Aminoalkyl methacrylate polymer and various beverages	Bitter taste and bitterness suppression	All sensors (8) of SA402B	Bitterness prediction from taste sensors (Channel 3 and Channel 4) and correlation of sensors data with human gustatory sensation test for a commercial dry syrup formulation of clarithromycin.	(Tanigake et al., 2003)

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Table 11 (*continued*)

Active Ingredient/Taste Substance	Formulation	Taste Masking Agent	Taste Assessed	Taste Sensor and Equipment/Machine	Remarks	Ref.
Clarithromycin, erythromycin, cefdinil, doxycycline, vancomycin, tetracycline, minocycline, oxytetracycline and bacampicillin	Commercial syrup, drug solutions and suspensions	Aminoalkyl methacrylate polymer	Bitterness and bitterness suppression	All sensors (8) of SA402B	Successful bitterness prediction from taste sensors (Channel 2 and Channel 3) and correlation of sensors data with human gustatory sensation test for a dry syrup product.	(Uchida et al., 2003)
L-leucine, L-isoleucine, L-valine, L-tryptophan, L-phenylalanine	Solutions of multiple, binary and single amino acids	None	Bitter, sour, astringency	All sensors of SA402	Relative sensor output values and CPA based PCA method for assessing and predicting bitterness was developed.	(Miyanaga et al., 2002)
Quinine HCl	Solutions	Sucrose, aspartame, tannic acid, phosphatidic acid and NaCl	Bitterness and bitterness suppression	Channel 4 of SA402	Taste sensor was used to predict bitterness suppression of substances and correlated with human taste panel studies.	(Nakamura et al., 2002)
Amitriptyline hydrochloride, D-chlorpheniramine maleate, dextromethorphan hydrobromide, dibucaine hydrochloride, diltiazem hydrochloride, imipramine hydrochloride, promethazine hydrochloride, propranolol hydrochloride, trimebutin maleate, quinine hydrochloride	Solutions	Sucrose and aspartame	Bitter taste	SA402	Multiple regression analysis was employed on data from taste sensor to quantify bitterness and obtain bitterness scores. A good correlation between bitterness score from taste sensors (Channels 2 and 4) and human taste panel studies was achieved.	(Uchida et al., 2001)
Quinine HCl, L-tryptophan, $MgCl_2$	Solutions	Phospholipid (BMI-60)	Bitterness and bitterness suppression	All sensors of SA 401	Disagreement of taste sensor data and taste panel studies for bitterness suppression warranted for further development of sensitive taste sensing membrane.	(Takagi et al., 2001)

in PVC membranes, viz. benzyl dimethyltetradecylammonium chloride and dodecyltrimethylammonium bromide as positively charged lipids and palmitic acid and stearic acid as neutral lipids, were fabricated and tested for sourness in acetic acid, citric acid and hydrochloric acid solutions (Szpakowska and Marja, 2009). ASSE were prepared by coating glassy carbon/poly(3,4-ethylenedioxythiophene) electrodes with lipids-PVC membrane. These taste sensors with ISE and ASSE were successfully demonstrated for discriminating different soft drinks containing lipophilic compounds could be successfully applied for discriminating different compositions of soft drinks. Data obtained from five ASSE containing different lipid and a Ag/AgCl reference electrode based sensor was processed through Principal Component Analysis (PCA) and Agglomerative Hierarchical Clustering (AHC) analysis for assessing fourteen different sensory attributes including taste, aftertaste and other attributes of three different beverages (Marjanska and Szpakowska, 2020).

Ciosek and coworkers designed a lipid membrane based sensor array which comprised 17 electrodes (Ciosek et al., 2004). Lipids, viz Tridodecylmethylammonium chloride, tetrakis(decyl)ammonium bromide, and bis(1-butylpentyl)adipate, a lipophilic ion potassium tetrakis[3, 5-bis(trifluoromethyl)phenyl]borate and plasticizers bis(2-ethylhexyl) sebacate 2-nitrophenyl octyl ether were used in fabricating sensing membranes. Three layered back propagation neural network based reduction of number of sensors to nine sensors was achieved for better discrimination and qualitative analysis of juice and water samples. This solid-state miniaturized potentiometric sensor was elaborated as a flow–through electronic tongue for electronic image analysis of particulates in dissolution studies. Dissolution studies of valsartan minitablets coated with Eudragit E were performed with this sensor to investigate the mechanisms of drug release and efficiency of bitterness suppression (Immohr et al., 2017; Wesoły et al., 2016). Further, extent of dissolution and image analysis of particles in dissolution medium were also studied for microspheres/microcapsules and physical mixtures of drugs and excipients (Jańczyk et al., 2010; Wesoły et al., 2017a).

A bitter database, Bitter DB ((http://bitterdb.agri.huji.ac.il/dbbitter.php), may be used to screen bitter substances which can serve as ligands for bitter taste receptors (Wiener et al., 2012). Docking simulations of bitter substances and taste masking agents may be performed to investigate electrostatic and hydrophobic interactions as the mechanism of bitterness suppression (Okuno et al., 2020). In one such study, docking-simulation study, surface plasmon resonance, AN0 taste sensor of SA402B responses and human gustatory sensation tests were used to examine the effect of dipeptides and amino acids as bitterness suppressing agents for dipenhydramine HCl. Simulations of bitter taste receptor (hTAS2R14) with dipenhydramine HCl, Glu-Glu and Asp-Asp revealed interactions among ligands and amino acids.

However, various types of taste assessment approaches are used for different lipids containing dosage forms in several research works and patents, which are enlisted in Table 11.

3.3. Commercial taste sensing systems based on LMTS

First commercial electronic tongue developed by Toko and coworkers at Kyushu University was marketed initially by Anritsu Corporation in 1993 and later from 2002 onwards by Intelligent Sensor Technology (Insent, a spin-off company of Anritsu Co.). A taste sensing system SA401 (Anritsu Co., Ltd) equipped with 7 taste sensors was the first electronic lipid membrane taste sensor commercialized in 1993 in Japan (http://www.insent.co.jp/en/products/sensorhistory_index.html). This taste sensing system was further developed further for a method used to assess change in the membrane potential due by adsorption (CPA method) of taste substances on lipid membranes and

Table 12
Composition of taste sensors of TS-5000Z.

Taste Sensor	Taste	Target Taste Substances	Lipid	Plasticizer
CT0	Saltiness	Dietary salts	TDAB, 1 hexadecanol	DOPP
CA0	Sourness	Tartaric acid, citric acid	PAEE, oleic acid, TOMA	DOPP
AAE	Umami and richness of umami substances	Amino acids, nucleic acids, soup, meat, stock sauce	TOMA	DOPP
C00	Initial bitterness of acidic substances, bitter aftertaste	Stock sauce, soup (Initial bitterness) and beer, coffee (bitter aftertaste)	TDAB	NPOE
BT0	Bitter aftertaste of HCl salts of drugs in medicines	HCl salts of drugs	PADE	BBPA
AN0	Initial bitterness of basic substances and bitter aftertaste	Quinine HCl, famotidine	PADE	DOPP
AE1	Initial astringency and astringent aftertaste	Wine, tea	TDAB	DOPP
GL1	Sweetness	Sweets, drinks	TDAB, trimellitic acid	DOPP
AC0	Aftertaste of basic bitter substances	Quinine HCl, famotidine	Hexadecanoic acid	DOPP
Prototype Model	Sweetness sensor for negatively charged artificial sweeteners	Acesulfame potassium	TDAB	NPOE
Prototype Model	Sweetness sensor for positively charged artificial sweetener	Aspartame	PAEE	BEO

Abbreviations: TOMA: Trioctylmetylammomium chloride; DOPP: Dioctyl phenylphosphonate; NPOE: 2-Nitrophenyl octyl ether; BEO: Butoxyethyl oleate; BBPA: bis(1-butylpentyl) adipate; TDAB: Tetradodecylammonium bromide; PADE; Phosphoric acid di-*n*-decyl ester; PAEE: phosphoric acid di(2-ethylhexyl) ester ; DOPP: dioctyl phenyl phosphonate.
References: Yassura et al., 2014, Liu et al., 2020, Toko et al., 2016 (Book Chapter); Kobayashi et al., 2010; Wu et al., 2020a, 2020b and http://www.insent.co.jp/en/products/taste_sensor_index.html#taste_information.

consequently a new version SA402 for objective taste assessment appeared in Japanese market in 1996. SA402 had an improved graphical user interface and customized to run on Windows 95. Later in 2000 SA402B model was launched for its compatibility to newer versions of Windows (98, ME, 2000, XP, 7). A new taste assessment method referred to as taste information method was a salient feature of this sensor. SA402B was marketed overseas as a taste sensor with global selectivity. Global selectivity was achieved by optimizing the hydrophobicity of the lipid membranes of the taste sensor. Substantial amount of lipids and plasticizer of adequate hydrophobicity contribute to selectivity and specificity for the assessment of bitter taste (Kobayashi et al., 2010). Sensors devoid of lipids do not respond to bitter stimuli. The latest version TS-5000Z (Insent Inc., Kangawa, Japan) of this taste sensing system for laboratories and quality control sections was introduced in 2007. TS-5000Z is equipped with three reference electrodes, three bitterness sensors (bitterness of cationic and neutral substances, bitterness of anionic substances, and aftertaste from hydrochloride salts), and one sensor for each sweetness, saltiness, umami stimuli, sourness and astringency. Chemical components of each taste sensor are presented in Table 12. TS-5000Z runs on a Linux operating system based network system capable of collecting and storage of data in a database which may provide access to multiple users via web browser. TS-5000Z is a biomimetic taste sensing system capable of assessing five basic tastes and 11 different kinds of taste information, viz. five basic tastes, astringency, richness or continuity of umami substance and aftertastes from acidic bitterness, basic bitterness, hydrochloride salts and astringency.

4. Conclusion

Lipids are indispensable part in case of taste masking and taste assessment of pharmaceutical formulations. Plastic flow and characteristic yield point makes lipids good candidates for tableting via compression processes. Lipids generate free fatty acids for which dissolution, bioavailability and absorption of drugs are highly revamped. Although different processes are applied for taste masking by lipids, hot-melt extrusion is probably the most straight forward method which eliminate the use of solvent and thereby drying step is also eliminated. Glyceryl esters and stearic acid are highly explored lipids for taste masking due to their suitable melting range and degree of unsaturation. Stearic acid is cheaper than glyceryl esters which makes it popular for its pilot scale use. LMTS have reached globally for their use in taste masked formulation development. LMTS based taste sensing system coupled with multivariate statistical analysis have grown immensely for optimizing taste masked formulations. LMTS have also established themselves as a great tool for chemical image analysis for characterizing pharmaceutical formulations which we could see a potential area to develop further in future.

Declaration of Competing Interest

The authors declare that there are no any conflicts of interest

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